

Analysis of Covid-19 chest x-rays

Data Science Project Report

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Chapter 1

Introduction

1.1 Background and Motivation

COVID-19 is a respiratory disease caused by the SARS-CoV-2 virus. During the pandemic, rapid and reliable diagnosis was an important challenge. Chest X-ray analysis has been explored as a possible tool to support diagnosis because abnormalities in lung structures can appear in infected patients.

A large number of studies have reported convolutional neural networks that classify COVID-19 from chest radiographs with high accuracy. Such results are encouraging, but they rest on an assumption that deserves scrutiny. Because COVID-19 radiographs were scarce during the early pandemic, public datasets were typically assembled by combining images from several repositories, with different diagnostic classes drawn from different sources. Where source and label are correlated, a classifier can reach high accuracy by recognising the acquisition conditions of a repository rather than the pathology in the lungs.

This concern motivates the structure of the present project. Classification performance alone cannot distinguish a model that reads pulmonary pathology from one that identifies its data source, so performance figures are reported here alongside experiments designed to establish what the models are actually using.

1.2 Related Work

El Houby (2024) evaluated frozen VGG19 and EfficientNetB0 backbones on the COVID-19 Radiography Database under several image enhancement schemes. On a balanced four-class subsample of 3,200 images, the best model reached 0.935 accuracy and 0.935 F1; on binary COVID-19 versus Normal classification it reached 95% accuracy. That study also compared full radiographs against lung-segmented inputs and found segmentation to reduce performance, the best segmented model reaching 91% on the binary task. The author suggested that full images may contain details outside the lung region that contribute to classification. The experiments in Chapter 6 examine this possibility directly, and suggest that such details reflect the source of an image rather than its pathology.

A parallel line of work questions whether such figures measure diagnostic ability at all. DeGrave, Janizek, and Lee (2021) trained models representative of published COVID-19 radiograph classifiers, on data reproducing that used in earlier studies, and

used explainable-AI techniques to show that they rely on confounding factors rather than pathology: the models appear accurate on held-out data from the same sources but fail on radiographs from new hospitals. The authors attribute this to the way training data was assembled, and further show that evaluation on external data is not by itself sufficient, since learned shortcuts may not impair performance in new hospitals.

The present project relates to this literature in two ways. Chapters 4 and 5 reach performance comparable to that reported by El Houby (2024), using the full deduplicated dataset rather than a balanced subsample. Chapter 6 then extends the shortcut-learning analysis in two respects: it traces the effect to a source-label correspondence documented in the dataset metadata, and it shows that the effect persists at input resolutions far below the scale at which radiographic findings are visible.

1.3 Problem Statement

This project investigates the automated classification of chest X-ray images into four diagnostic categories: COVID-19, Normal, Lung Opacity, and Viral Pneumonia. Chest X-ray imaging is widely used as a fast and accessible diagnostic tool for respiratory diseases, making automated image classification a valuable application of deep learning. The dataset consists of publicly available chest X-ray images collected from several repositories and includes corresponding lung segmentation masks. Details of the dataset and its characteristics are presented in Chapter 2.

The project addresses two questions:

1. **Performance.** What classification accuracy can convolutional neural networks achieve on this dataset, and how much does transfer learning from a pretrained backbone improve on a network trained from scratch?
2. **Attribution.** Which regions and properties of the input do these models rely on, and to what extent is their accuracy attributable to pulmonary pathology rather than to characteristics of the source repositories?

The second question is addressed through gradient-based attention analysis (Grad-CAM) applied to individual models, and through a series of ablations in which the information available to the model is restricted: to the lung field only, to the non-lung regions only, and to inputs downsampled far below the resolution at which radiographic findings remain visible.

The project focuses on image classification and does not aim to provide a medical diagnosis or replace healthcare professionals. Furthermore, the objective is not to localize abnormalities within the images.

1.4 Structure of this Report

Chapter 2 describes the dataset and reports an exploratory analysis of its composition, integrity, and intensity characteristics. Chapter 3 sets out the preprocessing pipeline and the measures taken to prevent data leakage. Chapter 4 establishes baseline performance using a small convolutional network trained from scratch, and Chapter 5 evaluates transfer learning from an EfficientNetB0 backbone, including fine-tuning and interpretability

analysis. Chapter 6 brings together evidence from the dataset metadata and from ablation experiments across all architectures to characterise what the models are learning. Chapter 7 summarises the findings and their implications.

Chapter 2

Dataset

2.1 Overview

The dataset consists of chest X-ray (CXR) images belonging to four diagnostic categories: *COVID-19*, *Normal*, *Lung Opacity*, and *Viral Pneumonia*. All images are provided in PNG format with a resolution of 299×299 pixels. In addition, lung segmentation masks are available for every image. The dataset was compiled from several publicly available repositories and published studies, with detailed source information provided in the accompanying metadata.

According to the dataset documentation, the four classes were collected from different source repositories rather than a single unified acquisition process. Table 2.1 summarizes the origin of each class.

Table 2.1: Sources of the chest X-ray images for each class.

Class	Source(s)
COVID-19	PadChest; German Medical School; SIRM/GitHub/Kaggle/Twitter; another GitHub source
Normal	RSNA Pneumonia Detection Challenge (8,851) + Kaggle Chest X-ray Pneumonia Dataset (1,341)
Lung Opacity	RSNA Pneumonia Detection Challenge only
Viral Pneumonia	Kaggle Chest X-ray Pneumonia Dataset only

The COVID-19 class was created by aggregating confirmed COVID-19 chest X-ray images from multiple public repositories to increase the number of available samples. Normal images were obtained primarily from the RSNA Pneumonia Detection Challenge dataset, with additional normal cases collected from the Kaggle Chest X-ray Pneumonia dataset. Lung Opacity images originate exclusively from the RSNA dataset and represent non-COVID pulmonary opacities. Viral Pneumonia images were obtained exclusively from the Kaggle Chest X-ray Pneumonia dataset.

Although combining multiple repositories increases the diversity of the dataset, it also introduces potential biases resulting from differences in image acquisition protocols, imaging equipment, preprocessing pipelines, compression methods, image quality, and patient populations. In particular, all Lung Opacity images originate from the RSNA

dataset, whereas all Viral Pneumonia images are drawn exclusively from the Kaggle dataset. Consequently, a classifier may learn source-specific characteristics instead of disease-related radiographic patterns.

This issue is commonly referred to as *shortcut learning*, where a model exploits acquisition- or dataset-specific artifacts rather than clinically relevant pathological features. To ensure that the trained model focuses on meaningful anatomical regions, explainable artificial intelligence (XAI) techniques such as Grad-CAM should be employed to verify that predictions are based on disease-related radiographic features instead of scanner- or dataset-specific artifacts.

2.2 Dataset Composition and Class Balance

The dataset consists of 21,165 chest X-ray (CXR) images distributed across four diagnostic categories: *COVID-19*, *Lung Opacity*, *Normal*, and *Viral Pneumonia*. Each chest X-ray image has a corresponding lung segmentation mask, resulting in a one-to-one mapping between images and masks.

Table 2.2 summarizes the distribution of images across the four diagnostic classes. The *Normal* class is the largest, accounting for 48.15% of all images, followed by *Lung Opacity* (28.41%), *COVID-19* (17.08%), and *Viral Pneumonia* (6.35%).

Table 2.2: Distribution of chest X-ray images across diagnostic classes.

Class	Images	Percentage
COVID-19	3,616	17.08%
Lung Opacity	6,012	28.41%
Normal	10,192	48.15%
Viral Pneumonia	1,345	6.35%
Total	21,165	100.00%

Figure 2.1 illustrates the class distribution within the dataset.

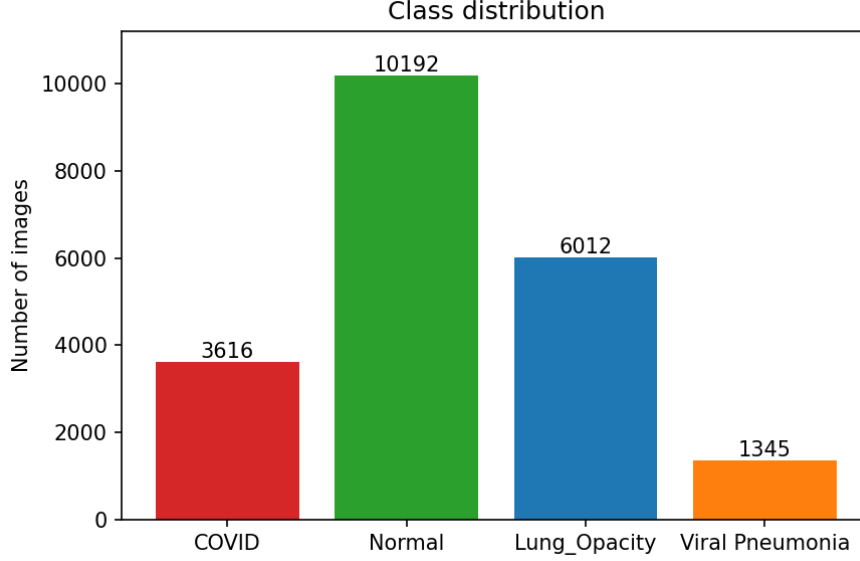


Figure 2.1: Distribution of chest X-ray images across the four diagnostic classes.

To determine whether the observed class frequencies differed significantly from an equal class distribution, a chi-square goodness-of-fit test was performed. The test indicated a highly significant deviation from uniformity ($\chi^2 = 8110.78$, $p < 0.001$). Furthermore, the effect size was large (Cohen’s $w = 0.619$), confirming the presence of substantial class imbalance.

The observed imbalance has important implications for model development and evaluation. During dataset partitioning, stratified sampling should be employed to preserve the original class proportions across the training, validation, and test sets. Furthermore, overall classification accuracy alone may provide a misleading assessment of model performance. Instead, evaluation should include class-sensitive metrics such as per-class precision, recall, macro-averaged F1-score, balanced accuracy, and the confusion matrix to ensure that performance is assessed fairly across all diagnostic categories.

2.3 Structural Integrity, Dimensions, and Image Encoding

To verify the integrity of the dataset, all chest X-ray images and lung segmentation masks were examined prior to model development. Every one of the 21,165 chest X-ray images had a corresponding segmentation mask with an identical filename, indicating a complete one-to-one correspondence between images and masks. Furthermore, no unmatched image-mask pairs or unreadable PNG files were identified.

The average X-ray file size varied slightly across the four diagnostic classes. The mean file sizes were 35.00 KB for COVID-19, 34.40 KB for Lung Opacity, 36.39 KB for Normal, and 39.36 KB for Viral Pneumonia. The relatively larger mean file size observed for the Viral Pneumonia class is explained by the presence of RGB images within this class, which are substantially larger than their grayscale counterparts.

Table 2.3 summarizes the structural properties of the dataset.

Table 2.3: Structural properties of the chest X-ray images and segmentation masks.

Property	X-rays	Masks
Number of files	21,165	21,165
Dimensions	299×299 px	256×256 px
L mode	21,025	0
RGB mode	140	21,165

Although every image has a corresponding segmentation mask, the image and mask dimensions differ. All chest X-ray images have a resolution of 299×299 pixels, whereas all segmentation masks have a resolution of 256×256 pixels. Consequently, any pixel-wise operation involving both the X-ray image and its corresponding mask requires the mask to be resized or otherwise aligned with the image dimensions.

2.3.1 RGB Images in the Viral Pneumonia Class

A detailed inspection of image encoding revealed that all 140 RGB images are contained exclusively within the Viral Pneumonia class. These RGB images represent 10.41% of the Viral Pneumonia samples (140 of 1,345 images), while every image in the remaining three diagnostic classes is stored as a single-channel grayscale (L-mode) image. Consequently, image encoding is associated with the diagnostic class in the raw dataset.

Table 2.4 summarizes the characteristics of the Viral Pneumonia images according to image encoding.

Table 2.4: Comparison of grayscale (L) and RGB images within the Viral Pneumonia class.

Mode	n	Mean (KB)	SD	Median	Range (KB)
L	1,205	37.45	2.54	37.47	26.37–47.27
RGB	140	55.82	5.86	56.28	39.83–71.42

Figure 2.2 compares the mean PNG file sizes of grayscale and RGB images within the Viral Pneumonia class.

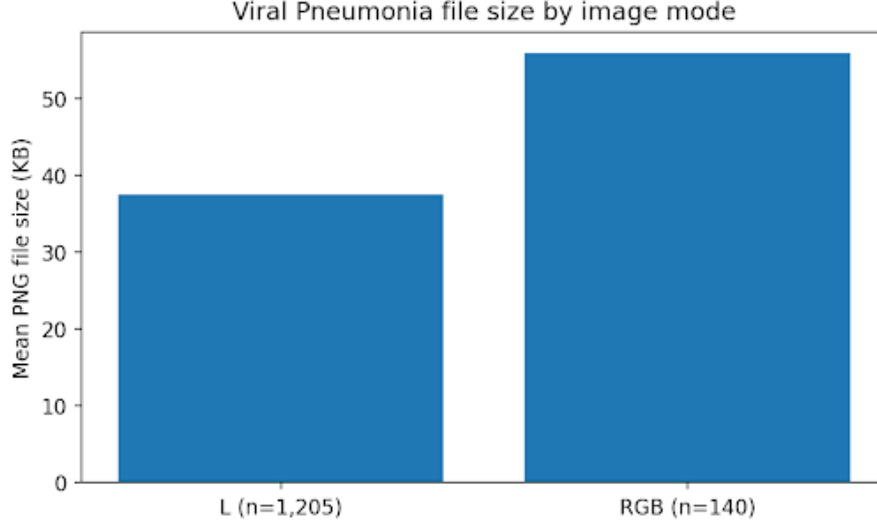


Figure 2.2: Mean PNG file size of Viral Pneumonia images stratified by image encoding.

As shown in Figure 2.2, RGB images are substantially larger than grayscale (L-mode) images within the same diagnostic category. This finding indicates that the elevated average file size observed for the Viral Pneumonia class is primarily attributable to differences in image encoding rather than disease-related image characteristics.

Since image encoding represents a technical characteristic rather than a clinically meaningful feature, all chest X-ray images should be converted to a common grayscale representation during preprocessing. Standardizing the image format eliminates a potential source of dataset bias and prevents the model from exploiting encoding-related differences as shortcut features.

2.4 Exact Duplicate Analysis

To identify redundant images, perceptual hashing (pHash) was applied using a Hamming-distance threshold of zero. This initial screening identified 287 candidate duplicate image pairs. A subsequent pixel-wise comparison confirmed that 91 of these pairs were exact pixel duplicates, whereas the remaining 196 pairs shared an identical perceptual hash but differed at the pixel level. All confirmed exact duplicates were found within the same diagnostic class, and no duplicate images were shared across different class labels.

Duplicate images were subsequently grouped into connected duplicate sets, and a single representative image was retained from each group. This process identified a total of 59 redundant files, corresponding to only 0.28% of the entire dataset. The distribution of redundant files across diagnostic classes is presented in Table 2.5.

Table 2.5: Redundant exact duplicate files identified after grouping connected duplicate sets.

Class	Redundant Files	Percentage of Class
COVID-19	51	1.41%
Lung Opacity	0	0.00%
Normal	1	0.01%
Viral Pneumonia	7	0.52%
Total	59	0.28%

Figure 2.3 illustrates the number of confirmed redundant images identified within each diagnostic class.

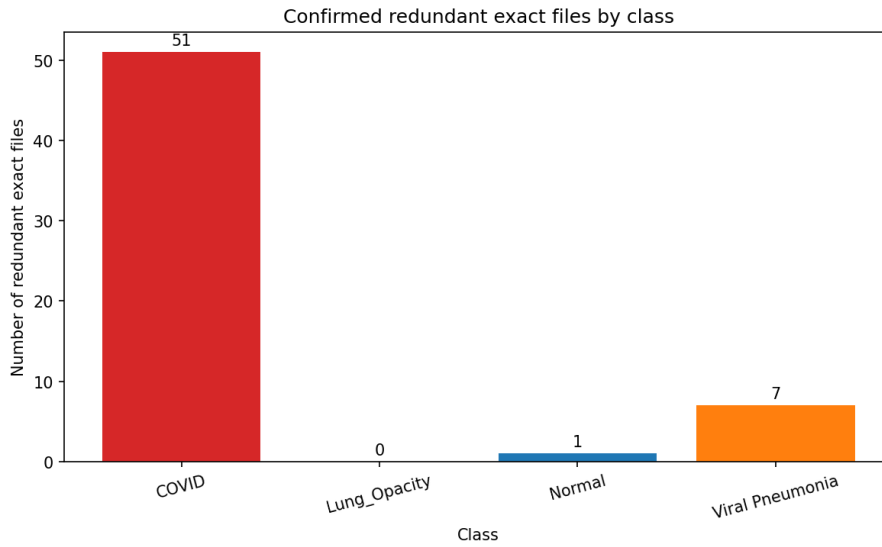


Figure 2.3: Distribution of confirmed redundant exact duplicate files by diagnostic class after grouping duplicate sets.

The identified duplicate images represent unnecessary redundancy within the dataset and, if retained, could introduce data leakage by allowing identical images to appear in both the training and evaluation sets. To prevent this issue, the 59 redundant files were removed prior to performing the training/validation/test split, while retaining one representative image from each exact-duplicate group. This preprocessing step helps ensure an unbiased evaluation of model performance and prevents overly optimistic performance estimates resulting from duplicate observations.

2.5 Whole-Image Brightness and Contrast

To characterize global image intensity, two descriptive features were calculated for each chest X-ray. *Brightness* was defined as the mean grayscale pixel intensity of the image, while *contrast* was represented by the standard deviation of grayscale pixel intensities within the image. To ensure consistency, all chest X-ray images were converted to grayscale prior to these calculations.

Table 2.6 summarizes the descriptive statistics for brightness and contrast across the four diagnostic classes. COVID-19 images exhibited the highest mean brightness (139.52), whereas Normal images showed the highest average global contrast (61.56). Lung Opacity and Viral Pneumonia displayed similar central brightness values, although their contrast differed slightly.

Table 2.6: Descriptive statistics of whole-image brightness and contrast by diagnostic class.

Class	Brightness			Contrast		
	Mean	SD	Median	Mean	SD	Median
COVID-19	139.52	25.03	142.59	54.65	14.83	54.05
Lung Opacity	126.02	23.57	120.82	57.40	9.81	56.97
Normal	129.39	22.45	125.51	61.56	9.55	63.53
Viral Pneumonia	125.37	18.98	126.86	58.75	10.78	59.71

Figure 2.4 presents the distributions of brightness and contrast for each diagnostic class. COVID-19 images tend to exhibit higher overall brightness, whereas the Normal class has the highest global contrast. The boxplots also reveal considerable within-class variability and numerous statistical outliers, particularly for COVID-19. Since these measurements were computed over the entire image, they may also be influenced by background regions outside the lungs. Consequently, evaluating brightness and contrast within the segmented lung regions may provide a more clinically meaningful assessment.

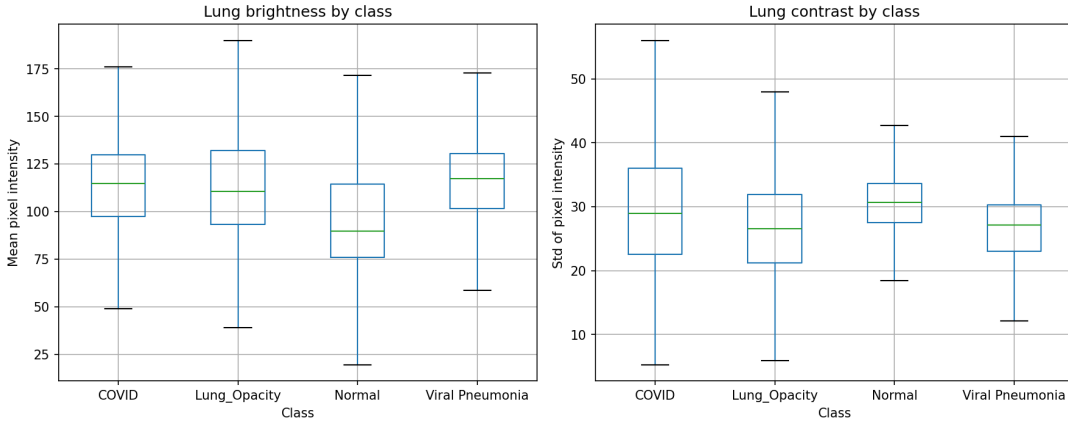


Figure 2.4: Whole-image brightness and contrast distributions by diagnostic class.

Figure 2.5 illustrates the relationship between whole-image brightness and contrast. Although class-specific tendencies are visible, substantial overlap exists among all four diagnostic classes. COVID-19 images exhibit a broader spread in both brightness and contrast, whereas Viral Pneumonia images are more tightly clustered. Normal images tend toward higher contrast, consistent with the descriptive statistics shown in Table 2.6. The considerable overlap indicates that global intensity characteristics alone are unlikely to provide sufficient discriminative information for reliable disease classification.

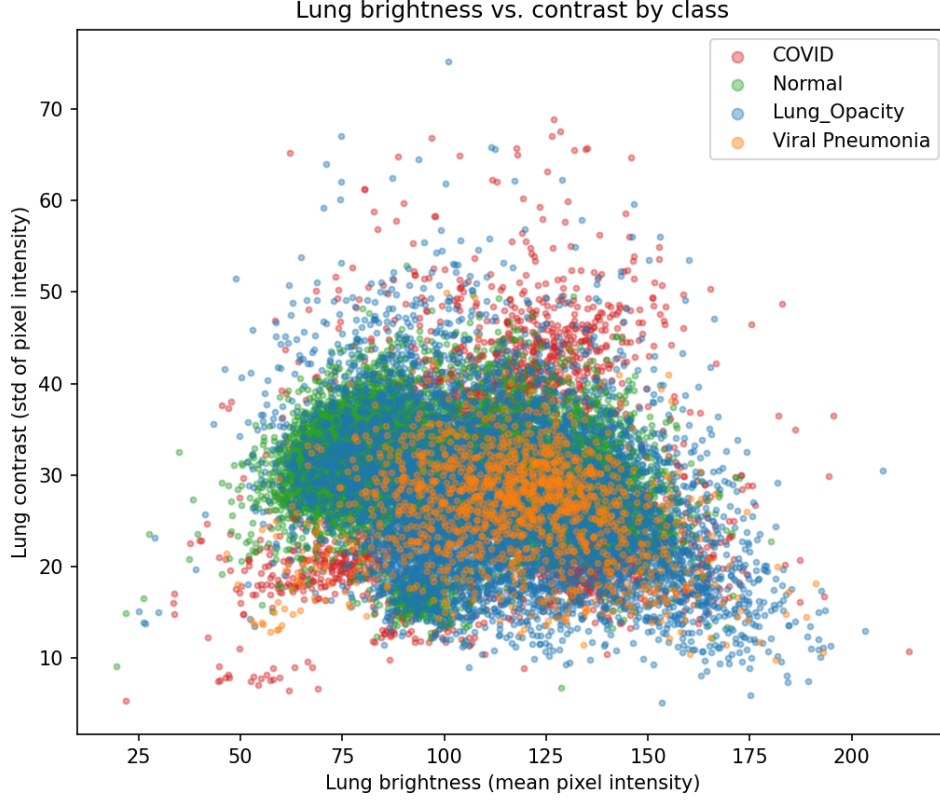


Figure 2.5: Relationship between whole-image brightness and contrast by diagnostic class.

The observed differences in global brightness and contrast are more likely to reflect variations in image acquisition protocols, scanner characteristics, and preprocessing procedures than disease-specific radiographic features. Consequently, these global intensity characteristics may represent technical confounding factors that a machine learning model could exploit as *shortcut features*. Applying intensity normalization during preprocessing and evaluating its impact on model performance can help determine whether the classifier relies on clinically meaningful radiographic patterns rather than global intensity differences.

2.5.1 Statistical Testing

Non-parametric statistical analyses were performed to compare brightness and contrast across the four diagnostic classes. A Kruskal–Wallis test indicated significant differences in whole-image brightness ($H(3) = 987.39$, $p < 0.001$), with a small overall effect size ($\varepsilon^2 = 0.0465$). Significant differences were also observed for whole-image contrast ($H(3) = 1393.78$, $p < 0.001$), with a modest effect size ($\varepsilon^2 = 0.0657$).

To quantify the magnitude of pairwise differences, Cohen’s d was computed for each pair of diagnostic classes. The resulting effect sizes are summarized in Table 2.7.

Table 2.7: Pairwise Cohen’s d effect sizes for whole-image brightness and contrast.

Comparison	Brightness	Contrast
COVID-19 vs Lung Opacity	0.559	-0.230
COVID-19 vs Normal	0.438	-0.618
COVID-19 vs Viral Pneumonia	0.601	-0.296
Lung Opacity vs Normal	-0.147	-0.431
Lung Opacity vs Viral Pneumonia	0.029	-0.135
Normal vs Viral Pneumonia	0.182	0.290

Post hoc Dunn tests with Bonferroni correction revealed significant brightness differences for every class pair except Lung Opacity versus Viral Pneumonia. Although the comparison between Normal and Viral Pneumonia remained statistically significant (adjusted $p = 0.0012$), the corresponding effect size was small ($d = 0.182$). For whole-image contrast, all six pairwise comparisons remained statistically significant following Bonferroni adjustment. The largest standardized difference was observed between COVID-19 and Normal ($d = -0.618$), indicating that COVID-19 images generally exhibit lower global contrast than Normal images.

Overall, these findings demonstrate that brightness and contrast differ systematically across diagnostic classes. However, the relatively small-to-moderate effect sizes and substantial overlap among class distributions suggest that global intensity characteristics alone are insufficient for robust disease classification. Instead, they should primarily be regarded as potential technical confounding variables that require careful preprocessing and interpretation during model development.

2.6 Summary of Current Evidence

Table 2.8 summarizes the principal findings obtained during the exploratory data analysis and their implications for the subsequent preprocessing and model development stages.

Table 2.8: Summary of key findings from the exploratory data analysis.

Domain	Main Result	Implication
Class balance	$\chi^2(3) = 8110.78$; Cohen's $w = 0.619$	Substantial class imbalance, with the Normal class dominating the dataset and Viral Pneumonia being underrepresented.
Integrity	0 unreadable files; 0 unmatched image-mask pairs	Complete correspondence between chest X-ray images and segmentation masks.
Image encoding	140 RGB X-rays, all belonging to the Viral Pneumonia class	Class-specific technical artifact; all images should be standardized to grayscale before model training.
Exact duplicates	59 redundant files, including 51 COVID-19 images	Duplicate images should be removed prior to dataset partitioning to prevent data leakage.
Lung-mask coverage	$H(3) = 1493.59$; $\varepsilon^2 = 0.0704$	Modest but statistically significant differences in lung-mask coverage across diagnostic classes.
Whole-image brightness	$H(3) = 987.39$; $\varepsilon^2 = 0.0465$	Small but significant class-associated differences in global image brightness.
Whole-image contrast	$H(3) = 1393.78$; $\varepsilon^2 = 0.0657$	Modest but significant class-associated differences in global image contrast.

Overall, the dataset is structurally complete and internally consistent. However, several characteristics were identified that could influence machine learning performance independently of clinically meaningful pathology. These include substantial class imbalance, class-specific RGB image encoding, concentrated exact duplication within the COVID-19 class, statistically significant differences in lung-mask coverage, and systematic differences in whole-image brightness and contrast across diagnostic categories.

Although these characteristics are not inherently problematic, they represent potential technical confounding factors that could encourage a model to learn dataset-specific shortcuts instead of disease-related radiographic features. Consequently, the preprocessing pipeline has been designed to address these issues through deduplication, image standardization, mask alignment, intensity normalization, and stratified dataset partitioning. The effectiveness of these preprocessing steps will be evaluated during model development to ensure that the resulting classifiers learn clinically meaningful representations and generalize reliably to previously unseen data.

Chapter 3

Preprocessing

The exploratory analysis in Chapter 2 identified several characteristics that could bias a classifier independently of pathology: exact duplicate images, class-specific RGB encoding, mismatched image-mask resolutions, class imbalance, and systematic differences in global brightness and contrast. The preprocessing strategy described here addresses these issues before model training while preserving the original dataset and documenting every decision for reproducibility.

3.1 Design Rationale

Each preprocessing choice follows directly from the exploratory findings. Duplicate removal prevents data leakage between training and evaluation sets. Grayscale conversion removes the class-associated RGB encoding found in 10.41% of Viral Pneumonia images. Mask resizing with nearest-neighbor interpolation preserves lung boundaries when aligning 256×256 masks with 299×299 X-rays. Stratified partitioning maintains the imbalanced class distribution across splits. Per-image min-max normalization reduces global intensity differences that may reflect acquisition protocol rather than pathology. Optional contrast enhancement (CLAHE), lung-region masking, and training-time augmentation are retained as controlled alternatives for later comparison.

Figure 3.1 summarizes the overall workflow from the raw collection of 21,165 images to the modeling-ready set of 21,106 samples.

Raw dataset (21,165 images) $\rightarrow$ remove 59 exact duplicates $\rightarrow$ cleaned dataset (21,106 images) $\rightarrow$ stratified split (70% / 15% / 15%) $\rightarrow$ per-image pipeline (grayscale, resize, optional CLAHE / lung mask / augmentation, min-max normalization) $\rightarrow$ training, validation, and test partitions
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Figure 3.1: End-to-end preprocessing workflow. CLAHE, lung masking, and augmentation were disabled in the baseline configuration but remain available for ablation studies.

3.2 Deduplication and Processed Dataset

Exact duplicate groups identified during exploration yielded 59 redundant files. These were excluded *before* splitting so that identical observations cannot appear in both training and evaluation subsets.

Table 3.1 reports the class distribution after removal. The relative proportions remain close to the raw dataset because duplicates represented only 0.28% of all images and were concentrated in the COVID-19 class, where 51 of 3,616 images (1.41%) were redundant.

Table 3.1: Class distribution after removing 59 exact duplicate files.

Class	Images	Percentage
COVID-19	3,565	16.89%
Lung Opacity	6,012	28.48%
Normal	10,191	48.29%
Viral Pneumonia	1,338	6.34%
Total	21,106	100.00%

The 196 image pairs that share a perceptual hash but differ at the pixel level were not removed automatically. They remain flagged for manual or similarity-based review because automatic exclusion could discard legitimately distinct acquisitions.

3.3 Stratified Partitioning

The cleaned dataset was divided using a two-step stratified procedure with a fixed random seed to ensure reproducibility:

1. 70% assigned to training and 30% held out for evaluation.
2. The holdout set divided equally into validation and test subsets, yielding an overall 70% / 15% / 15% partition.

Class proportions were preserved at each step. Table 3.2 lists the resulting sample counts; Table 3.3 shows that normalized class frequencies remain aligned across all three splits, with a maximum deviation of 0.03 percentage points for Viral Pneumonia.

Table 3.2: Absolute sample counts per split after deduplication and stratification.

Class	Train	Validation	Test	Total
COVID-19	2,495	535	535	3,565
Lung Opacity	4,208	902	902	6,012
Normal	7,134	1,528	1,529	10,191
Viral Pneumonia	937	201	200	1,338
Total	14,774	3,166	3,166	21,106

Table 3.3: Class proportions within each split.

Class	Train	Validation	Test
COVID-19	16.89%	16.90%	16.90%
Lung Opacity	28.48%	28.49%	28.49%
Normal	48.29%	48.26%	48.29%
Viral Pneumonia	6.34%	6.35%	6.32%

3.4 Per-Image Transformation

Each chest X-ray underwent the following sequence of transformations. The order matters: geometric operations precede contrast adjustment and masking, and normalization is applied last so that intensity scaling reflects the final pixel content presented to the model.

1. **Grayscale conversion.** All images were converted to single-channel grayscale, eliminating the RGB encoding bias described in Chapter 2.
2. **Resize.** Images were scaled to 224×224 pixels using area-based interpolation. Segmentation masks were resized to the same dimensions with nearest-neighbor interpolation to preserve binary lung boundaries.
3. **Optional augmentation.** When enabled, geometric and photometric transforms were applied to the image and mask jointly so that spatial correspondence is maintained (Section 3.5).
4. **Optional CLAHE.** Contrast Limited Adaptive Histogram Equalization (clip limit 2.0, tile size 8×8) was applied to the image only, targeting locally varying contrast without altering mask geometry.
5. **Optional lung masking.** Pixels outside the binarized lung region were zeroed before normalization, restricting intensity statistics to pulmonary tissue.
6. **Per-image min–max normalization.** Each image was scaled to $[0, 1]$ using its own minimum and maximum pixel value:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min} + \varepsilon}, \quad \varepsilon = 10^{-8}.$$

Independent normalization attenuates scanner- and acquisition-specific global brightness differences identified in the brightness–contrast analysis.

Figure 3.2 illustrates these stages on a representative COVID-19 chest X-ray. The aligned lung mask shows nearest-neighbor resizing from 299×299 to 224×224 . CLAHE and lung masking are shown for illustration; both were disabled in the baseline configuration.

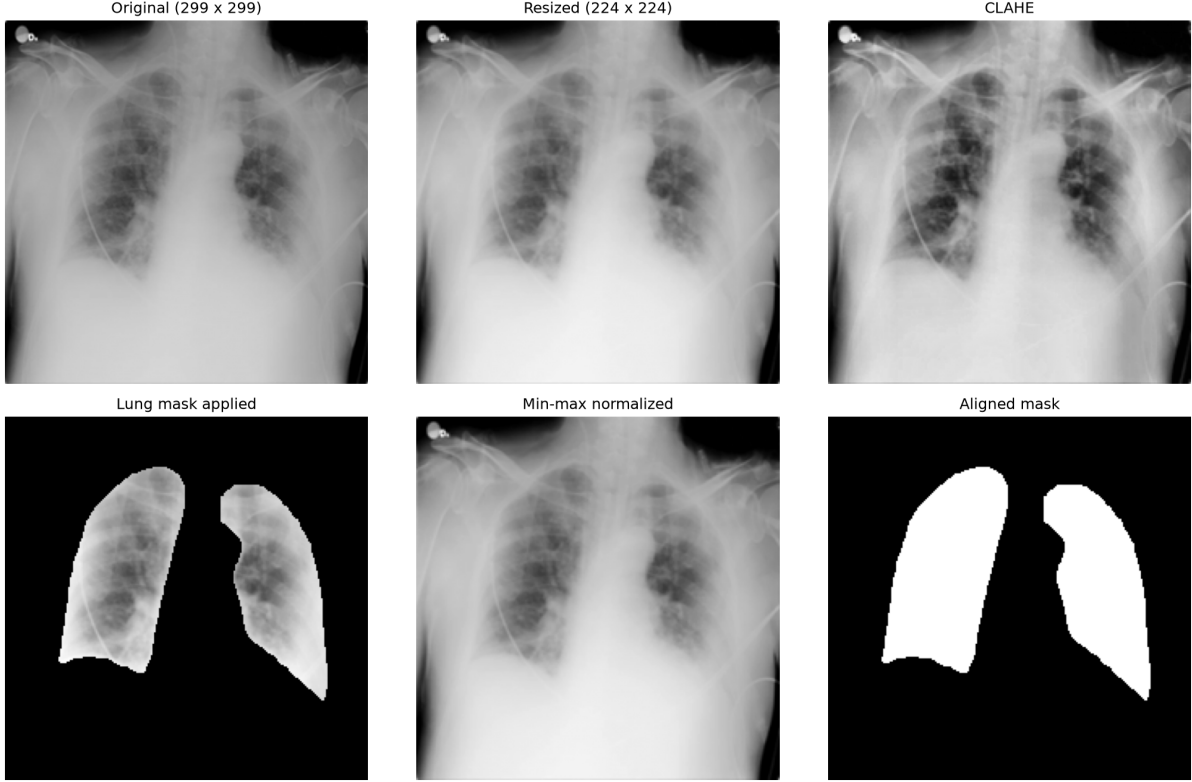


Figure 3.2: Per-image preprocessing stages on a representative COVID-19 chest X-ray: original grayscale image, resize to 224×224 , CLAHE, lung masking, min-max normalization, and aligned segmentation mask.

Table 3.4 summarizes the baseline configuration. CLAHE, lung masking, and augmentation were disabled so that their effect on downstream classification can be evaluated as controlled ablations.

Table 3.4: Baseline preprocessing configuration.

Setting	Baseline value
Output spatial size	224×224 pixels
Grayscale conversion	Applied to all images
CLAHE	Disabled
Lung mask application	Disabled
Lung mask binarization threshold	127
Training-time augmentation	Disabled
Random seed	42
Train / validation / test ratio	70% / 15% / 15%

3.5 Training-Time Augmentation

Data augmentation was designed to increase effective training-set diversity without breaking the correspondence between chest X-rays and lung segmentation masks. Geometric transforms were applied to both modalities jointly; photometric adjustments affected the

image only. Augmentation was restricted to the training partition so that validation and test performance reflect unmodified acquisitions.

Table 3.5 lists the operations and their activation probabilities. Each operation is applied stochastically and independently, so a single training sample may receive none, one, or several transforms.

Table 3.5: Augmentation operations and activation probabilities.

Operation	Probability	Effect
Horizontal flip	0.5	Mirror image and mask about the vertical axis
Rotation with zoom compensation	0.5	Up to $\pm 15^\circ$; borders reflected on image
Random zoom	0.5	Scale change up to $\pm 8\%$; center crop or pad
Translation	0.5	Shift up to $\pm 5\%$ of width and height
Brightness / contrast	0.4	Intensity scaling up to $\pm 10\%$; image only

Rotation without zoom compensation was evaluated separately and found to introduce empty border regions in the segmentation mask when reflect padding was used. The zoom-compensated rotation variant scales the image to fill the frame and uses zero padding on the mask, keeping lung labels binary. Figure 3.3 compares individual geometric transforms on a representative COVID-19 case. Figure 3.4 shows the combined effect of a single stochastic augmentation pass on the same sample.

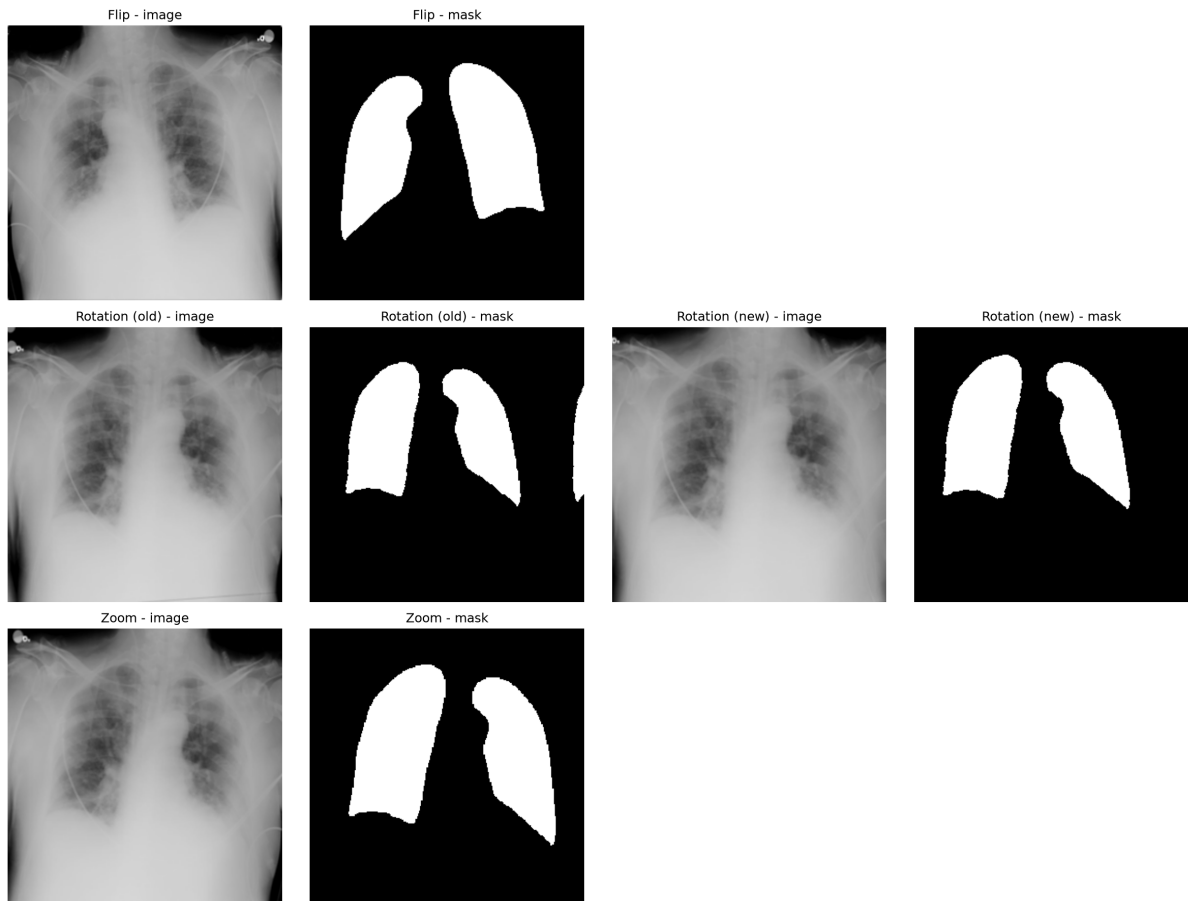


Figure 3.3: Individual augmentation operations on a representative COVID-19 chest X-ray: horizontal flip, rotation without zoom compensation, rotation with zoom compensation, and random zoom. Each row pairs the transformed image with its aligned lung mask.

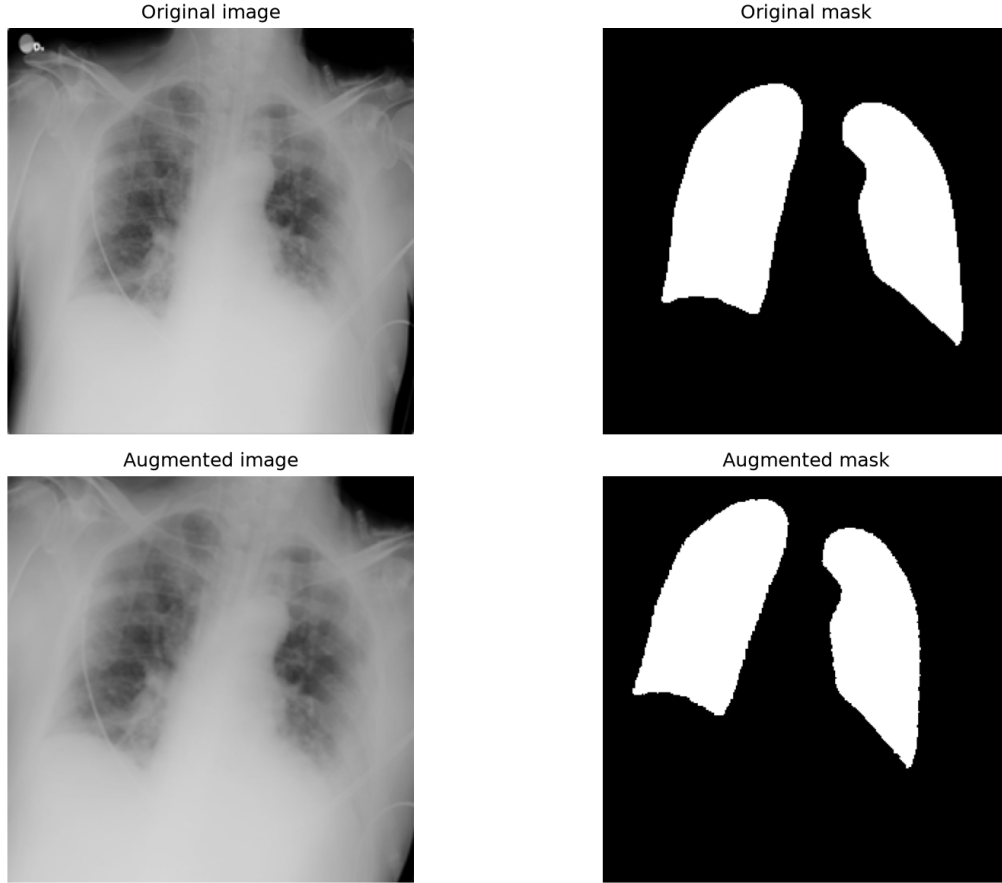


Figure 3.4: Original COVID-19 chest X-ray and lung mask compared with one stochastic augmentation pass (fixed random seed for reproducibility).

3.6 Verification and Reproducibility

The preprocessing workflow was verified at three levels before model training.

First, **transform correctness** was confirmed on synthetic images: normalized outputs lie in $[0, 1]$, repeated processing without augmentation yields identical results, lung masking zeroes non-pulmonary regions, and CLAHE changes pixel values while preserving spatial dimensions. Geometric augmentation preserves image and mask shape across 25 consecutive stochastic draws.

Second, **split integrity** was confirmed: the three partitions are disjoint, together they account for all 21,106 retained samples, and class proportions match the full dataset within 0.05 percentage points.

Third, **end-to-end consistency** was confirmed on the full dataset: all 21,106 images were read successfully with zero failures; processed image counts, split totals, and stacked sample counts all equal 21,106; and a complete rebuild of the baseline configuration produced bit-identical outputs to the saved training, validation, and test partitions ($14,774 + 3,166 + 3,166 = 21,106$).

A total of 78 automated checks covering transforms, augmentation, splitting, and pipeline regression passed. Stochastic steps used random seed 42 throughout.

3.7 Implications for Model Development

This preprocessing stage establishes a fixed, auditable baseline for convolutional network training. Stratified splitting and duplicate removal reduce leakage risk; grayscale standardization and per-image normalization target the technical shortcuts documented in Chapter 2. Because class imbalance persists after splitting—Viral Pneumonia remains at 6.34% of the training set—downstream evaluation must report per-class precision, recall, macro-averaged F1, balanced accuracy, and confusion matrices rather than overall accuracy alone.

Optional CLAHE, lung masking, and augmentation remain available for systematic comparison in subsequent experiments. Their impact on classification performance and on whether the model attends to pulmonary rather than acquisition-related features will be assessed during model development. A further variant, in which the lung masks are used only to locate a square region of interest (ROI) rather than to mask the image, is introduced in Section 4.3 where it is first used.

Chapter 4

Baseline Models

Two baseline configurations were established before evaluating more advanced approaches such as data augmentation and transfer learning. The objective was not only to obtain reference classification performance, but also to examine whether the convolutional neural network (CNN) relied on non-pulmonary or dataset-specific visual cues.

The first baseline used the complete chest radiograph and represented an unconstrained classification setting. However, additional region-based experiments showed that substantial predictive information was also present outside the segmented lung fields. Therefore, a second baseline was introduced using a lung-centred region of interest (ROI). In this configuration, the lung segmentation masks were used only to localise the thoracic ROI and were not applied directly to the input image. This avoided the artificial black boundaries introduced by hard lung masking while reducing peripheral image content.

The two baselines therefore served complementary purposes:

- **Full-image baseline:** measures the classification performance achievable from the complete chest X-ray.
- **Lung-ROI baseline:** provides a more spatially controlled reference in which peripheral image regions are reduced while the original radiographic appearance is preserved.

Neither baseline used data augmentation or transfer learning.

4.1 Experimental Setup

Both baselines used the same deliberately simple CNN trained from scratch. The architecture consisted of three convolutional blocks followed by global average pooling and a four-class softmax classifier:

Simple CNN Architecture

Input: $128 \times 128 \times 1$
Conv2D, 16 filters $\rightarrow$ MaxPooling
Conv2D, 32 filters $\rightarrow$ MaxPooling
Conv2D, 64 filters $\rightarrow$ MaxPooling
Global Average Pooling
Dense layer with 4-class softmax output

The model contained 23,556 trainable parameters. The classification task included four classes: COVID, Lung Opacity, Normal, and Viral Pneumonia. The same deduplicated dataset, train/validation/test partitions, random seed, class-weighting strategy, and optimisation settings were maintained across both baseline experiments.

The class distributions were identical for the two baselines and are shown in Table 4.1.

Table 4.1: Class distribution used for the baseline experiments.

Class	Train	Validation	Test
COVID	16.89%	16.90%	16.90%
Lung Opacity	28.48%	28.49%	28.49%
Normal	48.29%	48.26%	48.29%
Viral Pneumonia	6.34%	6.35%	6.32%

Class weighting was used to reduce the effect of class imbalance. Training was performed for a maximum of 100 epochs with early stopping based on validation loss.

4.2 Baseline 1: Full-Image Simple CNN

The first baseline was trained directly on the complete chest radiograph, without augmentation or additional intensity or geometry normalisation.

Its test-set performance is presented in Table 4.2.

Table 4.2: Test-set performance of the full-image simple CNN baseline.

Class	Precision	Recall	F1-score
COVID	0.766	0.830	0.796
Lung Opacity	0.799	0.766	0.782
Normal	0.850	0.831	0.840
Viral Pneumonia	0.840	0.945	0.889
Macro average	0.813	0.843	0.827

The full-image model achieved a test accuracy of **81.96%** and a macro F1-score of **82.7%**. This represents the strongest raw baseline performance obtained with the simple CNN.

However, the full-image model should be interpreted as a *naive performance baseline*. A separate background-only experiment achieved approximately 73.40% test accuracy, indicating that non-lung regions contain substantial class-correlated information. Therefore, high full-image classification performance does not necessarily imply that the model relies exclusively on clinically relevant pulmonary features.

4.2.1 Grad-CAM Analysis of the Full-Image Baseline

Grad-CAM was applied to the full-image model to investigate the spatial regions associated with its predictions. Quantitative attention analysis was performed on the complete test set of 3,166 images.

The mean lung attention fraction was 20.7%, whereas the segmented lungs occupied approximately 24.0% of the image area. The resulting mean lung-attention enrichment was 0.868. An enrichment value of 1 indicates that attention is approximately proportional to lung area, whereas values below 1 indicate relative under-attention to the lung fields.

Class-specific results are reported in Table 4.3.

Table 4.3: Class-wise Grad-CAM lung attention for the full-image baseline.

Class	Lung Attention	Lung Area	Enrichment
COVID	24.46%	24.38%	1.063
Lung Opacity	11.63%	21.80%	0.518
Normal	25.24%	25.18%	1.024
Viral Pneumonia	16.81%	23.22%	0.731

The results indicate strongly class-dependent attention behaviour. COVID and Normal cases exhibited approximately proportional lung attention, whereas Lung Opacity and Viral Pneumonia showed substantially lower lung-attention enrichment.

Qualitative inspection further revealed heterogeneous attention patterns. Some predictions were associated with activations within the pulmonary fields, while others showed strong responses near image borders or outside the segmented lung regions. These findings suggest that the full-image CNN may use a combination of pulmonary information and non-pulmonary or dataset-specific visual cues.

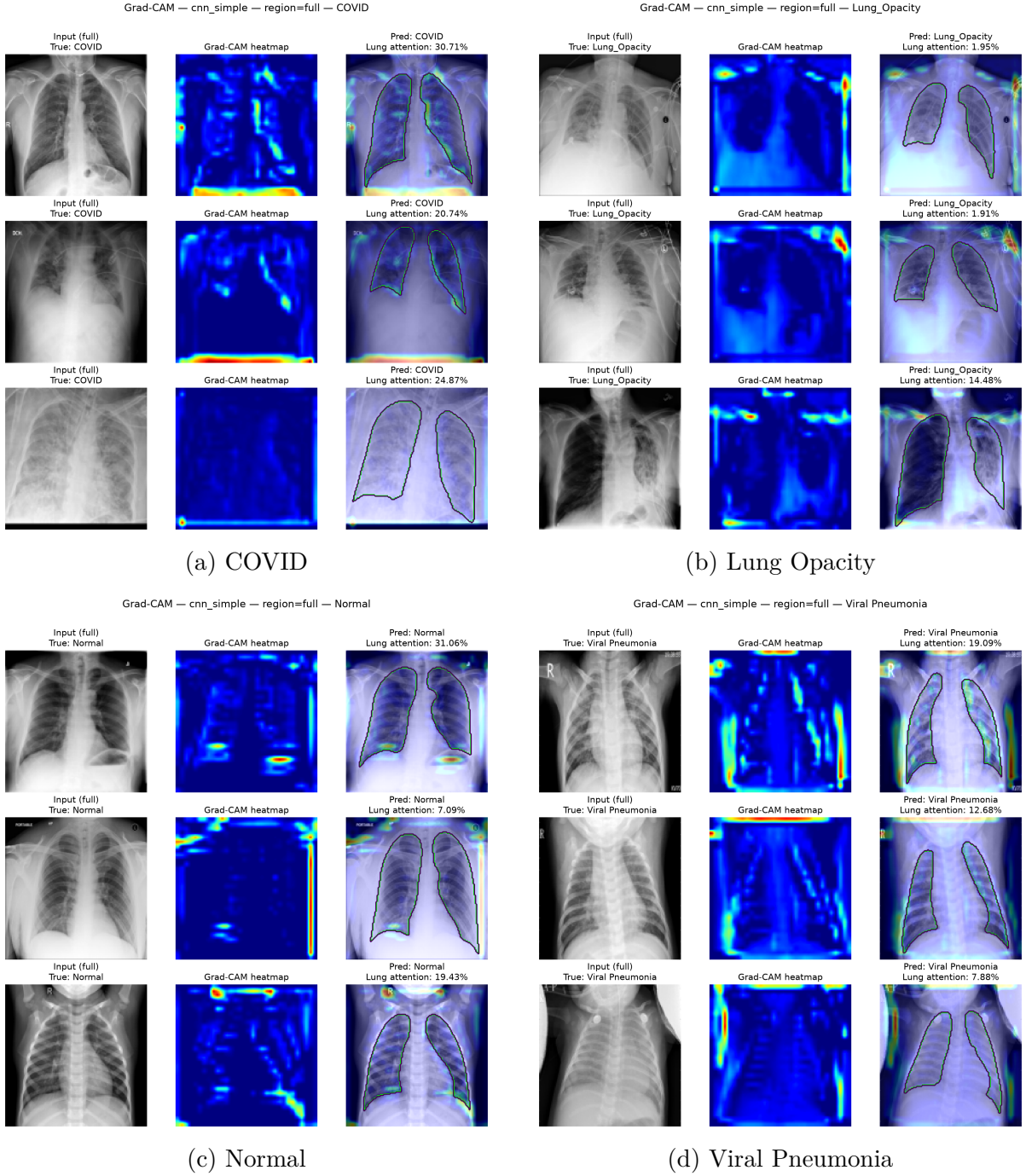


Figure 4.1: Representative Grad-CAM examples for the full lung baseline, grouped by class.

4.2.2 Motivation for a Lung-Centred ROI Baseline

A hard lung-masking experiment was initially used to restrict the CNN to the segmented pulmonary regions. This model achieved 72.93% test accuracy and 69.0% macro F1. However, hard masking introduced a sharp artificial boundary between the retained lung pixels and the zero-valued background.

Grad-CAM visualisations frequently showed activation around these segmentation boundaries. Consequently, the reduction in performance could not be attributed solely

to the removal of background information, because the masking operation itself changed the image representation and potentially introduced new geometric cues.

To obtain a cleaner controlled baseline, a lung-centred ROI preprocessing strategy was therefore introduced.

4.3 Baseline 2: Lung-ROI Simple CNN

For the second baseline, lung segmentation masks were used only to determine the spatial extent of the lungs. The image itself was not multiplied by the mask.

The preprocessing pipeline was:

1. Load the original chest X-ray and corresponding lung mask.
2. Retain only the two largest connected components of the mask, discarding small spurious regions.
3. Use the mask to determine the joint bounding box of the two lungs.
4. Add a small spatial margin around the lung region.
5. Construct a square ROI centred on the bounding box.
6. Where the square extends beyond the image boundary, pad with the median intensity of the image rather than cropping.
7. Preserve the original radiographic intensities and anatomical appearance.
8. Resize the ROI to 128×128 pixels.

This preprocessing reduced variation in overall framing and lung scale while avoiding hard segmentation boundaries. No rotation normalisation, intensity normalisation, or augmentation was applied.

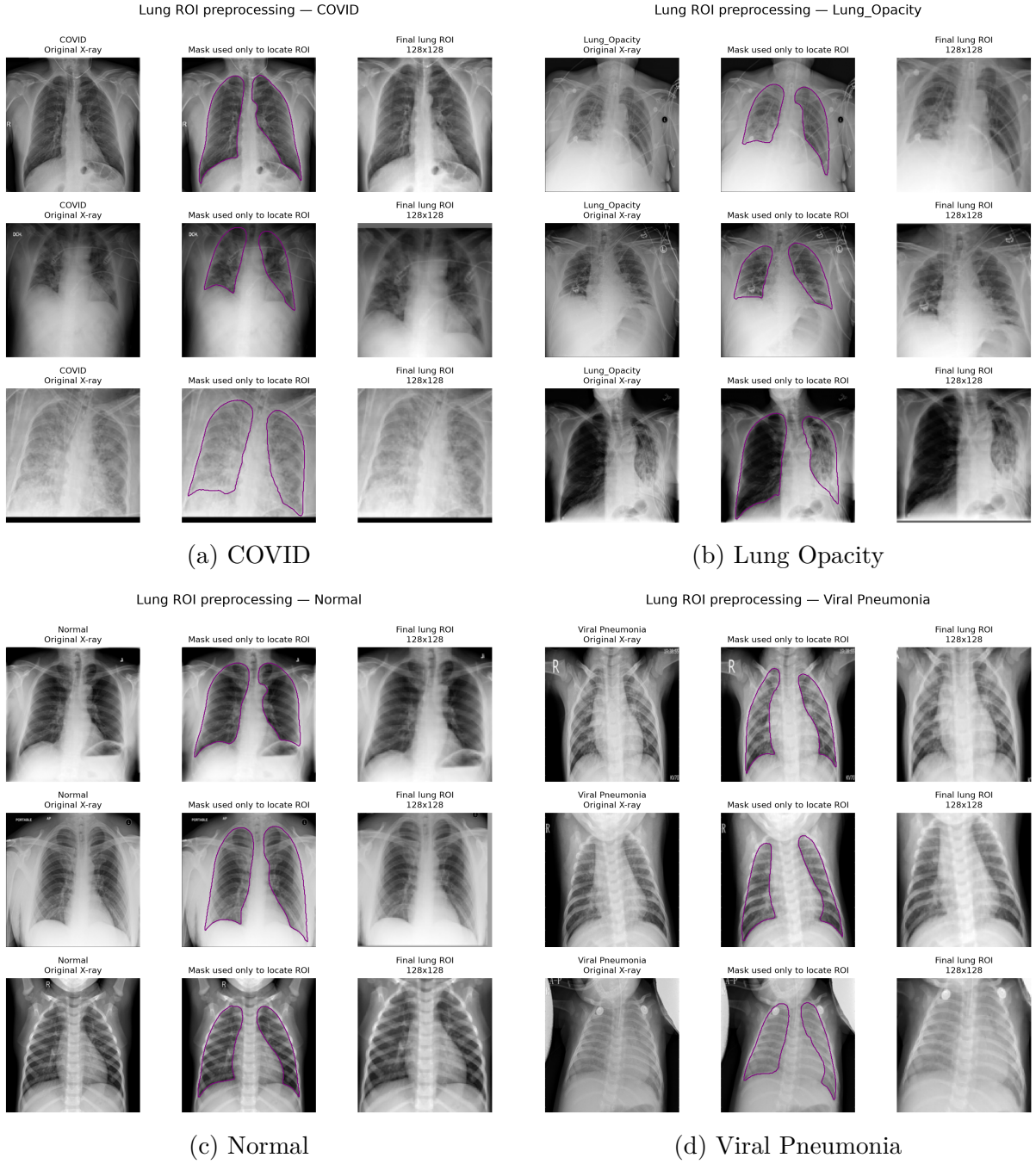


Figure 4.2: Class-wise examples of the lung-centred ROI preprocessing pipeline.

The lung-ROI model achieved the test-set performance shown in Table 4.4.

Table 4.4: Test-set performance of the lung-ROI simple CNN baseline.

Class	Precision	Recall	F1-score
COVID	0.649	0.703	0.675
Lung Opacity	0.822	0.691	0.751
Normal	0.828	0.853	0.840
Viral Pneumonia	0.746	0.940	0.832
Macro average	0.761	0.797	0.774

The lung-ROI model achieved **78.71% test accuracy** and **77.4% macro F1**. Its train, validation, and test accuracies were 79.86%, 78.74%, and 78.71%, respectively, indicating close agreement between the data partitions and little evidence of severe overfitting.

4.3.1 Grad-CAM Analysis of the Lung-ROI Baseline

Grad-CAM was also applied to the lung-ROI baseline. Unlike the hard-masked experiment, no artificial segmentation contour was present in the CNN input. Therefore, activation outside the segmented lung contour corresponds to actual anatomical or image regions contained within the ROI.

Qualitative inspection showed heterogeneous attention across the four classes. COVID examples generally demonstrated a greater degree of pulmonary activation, although attention was not exclusively confined to the lung fields. In several Lung Opacity and Viral Pneumonia examples, correctly classified images exhibited strong activation in peripheral, inferior, mediastinal, or ROI-border regions despite relatively low activation within the segmented pulmonary interior. Normal cases showed mixed behaviour, with both pulmonary and peripheral activation patterns.

These observations suggest that ROI cropping substantially reduces peripheral image information but does not completely prevent the CNN from exploiting extra-pulmonary or framing-related cues within the cropped thoracic region.

Because this ROI Grad-CAM assessment is currently based on representative qualitative examples rather than complete test-set quantitative attention analysis, the findings are interpreted as exploratory.

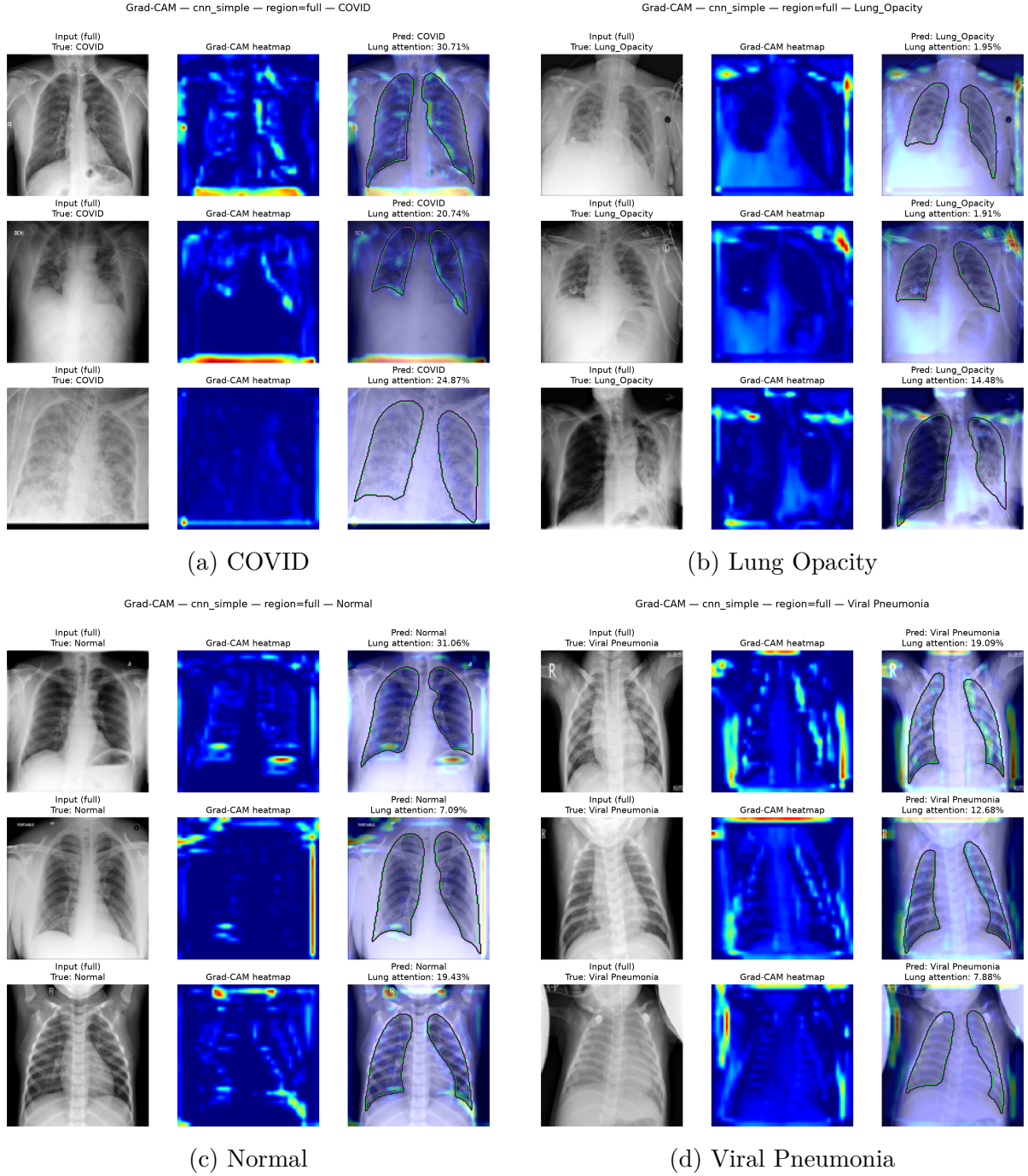


Figure 4.3: Representative Grad-CAM visualisations for the lung-ROI baseline, grouped by class. Although the ROI reduces peripheral image content and avoids hard segmentation boundaries, model attention remains heterogeneous and is not restricted exclusively to the pulmonary parenchyma.

4.3.2 Comparison of the Two Baselines

Table 4.5 compares the two principal baselines.

Table 4.5: Comparison of the two principal simple-CNN baselines.

Baseline	Test Accuracy	Macro F1
Full-image simple CNN	81.96%	82.7%
Lung-ROI simple CNN	78.71%	77.4%

Restricting the model to the lung-centred ROI reduced test accuracy by 3.25 percentage points and macro F1 by 5.3 percentage points. Nevertheless, the ROI model retained a substantial proportion of the full-image classification performance.

The class-wise recall comparison is shown in Table 4.6.

Table 4.6: Class-wise recall of the full-image and lung-ROI baselines.

Class	Full Image	Lung ROI
COVID	83.0%	70.3%
Lung Opacity	76.6%	69.1%
Normal	83.1%	85.3%
Viral Pneumonia	94.5%	94.0%

The largest reduction occurred for COVID classification. In contrast, Normal and Viral Pneumonia recall remained relatively stable.

These results suggest that peripheral image regions contribute to the performance of the full-image model, but they are not the sole source of discriminative information. A substantial level of classification performance remained when the input was restricted to a lung-centred thoracic ROI.

4.4 Baseline Interpretation and Role in Subsequent Experiments

The two baselines provide complementary reference points for subsequent model development.

The full-image model achieved the highest baseline performance, with 81.96% test accuracy and 82.7% macro F1. However, the background-only experiment and Grad-CAM analysis indicate that non-lung regions contain substantial class-correlated information. Consequently, the full-image model is retained as an unconstrained or naive performance baseline.

The lung-ROI model achieved 78.71% test accuracy and 77.4% macro F1 while substantially reducing peripheral image content and avoiding the hard-mask boundary artefact. It therefore serves as a more spatially controlled second baseline.

These two configurations were retained for the subsequent transfer-learning experiments. A practical difference is the input resolution: the simple CNN baselines use 128×128 images, whereas the transfer-learning models use 224×224 inputs. To assess whether this resolution difference alone could account for performance changes, the simple CNN was also evaluated at 224×224 . Increasing the input resolution did not improve performance. For full-image inputs, test accuracy decreased from 81.96% to 77.95%, while for lung-centered ROI inputs performance remained nearly unchanged (78.71% versus

78.40%). These results indicate that the low-capacity CNN did not systematically benefit from the higher input resolution, supporting the use of 128×128 as the baseline resolution.

This design supports two complementary questions:

1. Does a more advanced model improve overall classification performance on the complete chest radiograph?
2. Does the model also improve performance when access to obvious peripheral and background cues is reduced?

The second comparison is particularly important because an improvement in full-image accuracy alone does not necessarily imply improved use of clinically relevant pulmonary information.

Chapter 5

Transfer Learning

5.1 Transfer-Learning Approach and Experimental Setup

In addition to the two simple CNN baselines trained from scratch (Chapter 4), a transfer-learning approach was evaluated using an EfficientNetB0 backbone pretrained on ImageNet. The motivation was to test whether features learned on a large, general-purpose image dataset could improve classification performance on a relatively small, domain-specific chest radiograph dataset, and whether such a model relies on the same visual cues as the CNN trained from scratch.

The classification head placed on top of the backbone consisted of:

Transfer-Learning Head
EfficientNetB0 backbone (ImageNet weights)
Global Average Pooling
Dropout (rate 0.3)
Dense layer, 128 units, ReLU
Dropout (rate 0.3)
Dense layer, 4-class softmax output

Training used the same manifest, the same 70/15/15 stratified train, validation, and test split, and the same random seed (42) as the CNN baselines, so results are directly comparable to those reported in Chapter 4.

A two-phase training design was used throughout this chapter. In phase 1, the EfficientNetB0 backbone was kept frozen and only the classification head was trained. In several experiments (Section 5.4), a second phase then unfroze the backbone and continued training end-to-end at a substantially lower learning rate. This design allowed the new head to first adapt to the pretrained features before any backbone weights were disturbed, reducing the risk of destructive early updates.

5.2 Baseline Transfer Model

The first transfer-learning experiment kept the EfficientNetB0 backbone entirely frozen and trained only the classification head, without augmentation or class weighting. Its test-set performance is presented in Table 5.1.

Table 5.1: Test-set performance of the frozen EfficientNetB0 transfer model.

Class	Precision	Recall	F1-score
COVID	0.943	0.903	0.923
Lung Opacity	0.906	0.808	0.854
Normal	0.869	0.946	0.906
Viral Pneumonia	0.984	0.910	0.945
Macro average	0.925	0.892	0.907

The frozen transfer model achieved a test accuracy of **89.70%** and a macro F1-score of **90.7%**. This already exceeds the strongest simple-CNN baseline (the full-image model at 81.96% accuracy and 82.7% macro F1, Section 4.2) by approximately 7.7 accuracy points and 8.0 macro-F1 points, without any fine-tuning of the pretrained backbone. This result confirms that ImageNet-derived features transfer usefully to this chest radiograph classification task, even though the source and target domains are visually very different.

5.3 Optimisation: Augmentation, Class Weighting, and the Generalisation Gap

Following the initial baseline, several optimisation experiments were conducted to evaluate whether standard machine-learning tools for improving generalisation and handling class imbalance could improve on the frozen baseline, in line with the project’s Step 2 objective of using statistics and optimisation tools to fully exploit and understand the model’s results.

5.3.1 Data Augmentation and the Effect of Horizontal Flipping

An augmentation pipeline combining random horizontal flipping and small random rotations was first applied during training. Chest radiographs are not perfectly left-right symmetric in a clinically meaningless way, since heart position and other incidental laterality cues can appear in the images, so a second variant disabled horizontal flipping and used rotation only. Table 5.2 compares the two variants on the test split.

Table 5.2: Effect of horizontal flipping on the augmented transfer model (test split).

Model	Accuracy	Macro F1	COVID F1	Lung Opacity F1	Normal F1	Viral Pneumonia F1
Flip + rotation	87.62%	0.884	0.879	0.832	0.892	0.935
Rotation only	88.38%	0.896	0.883	0.850	0.895	0.956

Removing horizontal flipping improved every test-set metric, with the largest gains on Lung Opacity (+1.85 F1 points) and Viral Pneumonia (+2.07 F1 points). This is consistent with horizontal flipping introducing a mild, unhelpful distribution shift for this task rather than a useful invariance, and suggests that flipping should be treated as a tunable hyperparameter for chest radiographs rather than an unquestioned default.

5.3.2 Class Weighting

A separate experiment applied class weights inversely proportional to class frequency, without augmentation, to counter the dataset’s imbalance between the majority Normal and Lung Opacity classes and the minority COVID and Viral Pneumonia classes. The class-weighted model reached **88.34%** test accuracy and **0.894** macro F1, close to the augmented variants. Class weighting mainly redistributed performance toward the minority classes rather than closing the gap between training and test performance, indicating that it addresses a different problem (class imbalance) than augmentation (overfitting).

5.3.3 Generalisation Gap Across Runs

Table 5.3 reports the gap between train and test performance for the frozen baseline and its two augmentation variants, as a simple diagnostic of overfitting.

Table 5.3: Train-test generalisation gap for the frozen-backbone transfer models.

Model	Test Accuracy	Accuracy Gap	Macro-F1 Gap
Frozen baseline (no augmentation)	89.70%	2.77 pts	2.67 pts
Flip + rotation augmentation	87.62%	2.72 pts	2.66 pts
Rotation-only augmentation	88.38%	1.98 pts	1.27 pts

Rotation-only augmentation roughly halves the frozen baseline’s generalisation gap (1.98 versus 2.77 accuracy points, 1.27 versus 2.67 macro-F1 points) for a comparatively small reduction in raw test accuracy. This represents a genuine optimisation trade-off: a model that generalises somewhat better within this dataset, at a small cost in headline performance, although it does not on its own resolve the background-reliance concerns raised in Section 5.5.

5.4 Advanced Modelling: Fine-Tuning the Backbone

Following the project’s Step 3 objective of advanced modelling using deep learning, the EfficientNetB0 backbone was unfrozen and fine-tuned end-to-end after the initial frozen-head phase. Phase 1 trained the head only for 10 epochs; phase 2 then unfroze the entire backbone and continued training for 15 further epochs at a much lower learning rate (10^{-5} , versus 10^{-3} for the head-only phase), for 25 epochs in total.

Figure 5.1 shows the resulting training history. Validation loss briefly increases at the start of phase 2, a normal consequence of unfreezing a large pretrained backbone and updating it for the first time, before decreasing steadily to a final value substantially lower than at the end of phase 1 (from 0.310 to 0.189), with validation accuracy rising from 88.9% to 93.3% over the fine-tuning phase.

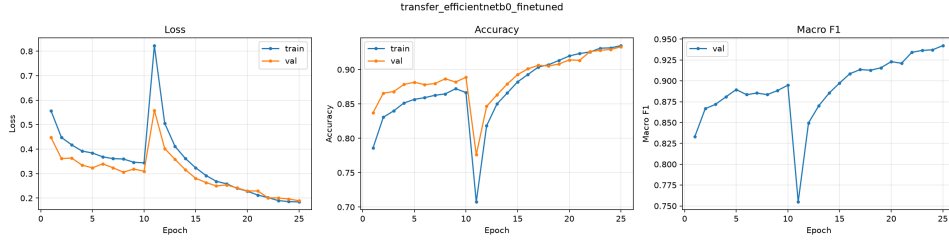


Figure 5.1: Training and validation curves for the fine-tuned EfficientNetB0 model, showing the frozen-head phase followed by end-to-end fine-tuning.

The fine-tuned model’s test-set performance is presented in Table 5.4.

Table 5.4: Test-set performance of the fine-tuned EfficientNetB0 transfer model.

Class	Precision	Recall	F1-score
COVID	0.931	0.978	0.954
Lung Opacity	0.958	0.814	0.880
Normal	0.898	0.962	0.929
Viral Pneumonia	0.980	0.980	0.980
Macro average	0.942	0.933	0.936

The fine-tuned model achieved a test accuracy of **92.36%** and a macro F1-score of **93.6%**, the strongest result obtained anywhere in this project. Fine-tuning improved every class relative to the frozen baseline except Lung Opacity, whose F1-score changed only marginally (0.854 to 0.880), and produced a particularly large gain in COVID recall (90.3% to 97.8%). The confusion matrix for this model is shown in Figure 5.2.

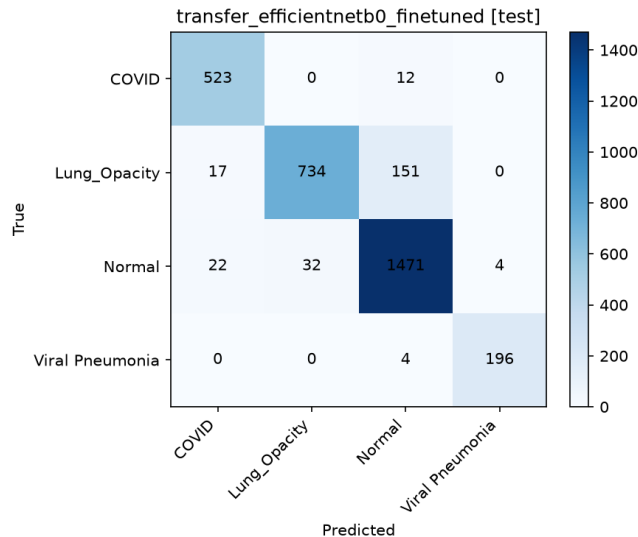


Figure 5.2: Test-set confusion matrix for the fine-tuned EfficientNetB0 model.

As with the full-image simple-CNN baseline, however, this headline performance must be interpreted alongside the interpretability analysis in Section 5.5 before it is read as evidence of genuinely improved pulmonary diagnosis.

5.5 Interpretability: Grad-CAM and the Lung-Focus Causal Analysis

Grad-CAM was applied to the transfer-learning models using the same lung segmentation masks employed for the CNN baselines’ interpretability analysis (Section 4.2.1). Beyond qualitative visualisation, a quantitative *lung-focus* metric was computed for every image: the share of each Grad-CAM heatmap’s total activation that falls inside the segmented lung mask. This is compared against a *chance* level, defined as the lung mask’s own share of the image area, that is, the score a spatially uninformative model would obtain by construction. A model that attends to the lungs no more than chance provides no evidence that it is using pulmonary information specifically.

5.5.1 Qualitative and Quantitative Lung Focus of the Frozen Model

Figure 5.3 shows representative Grad-CAM examples for the frozen transfer model, with the lung boundary overlaid in green. Heatmaps frequently activate at chest borders, shoulders, or corners rather than tightly within the lung fields.

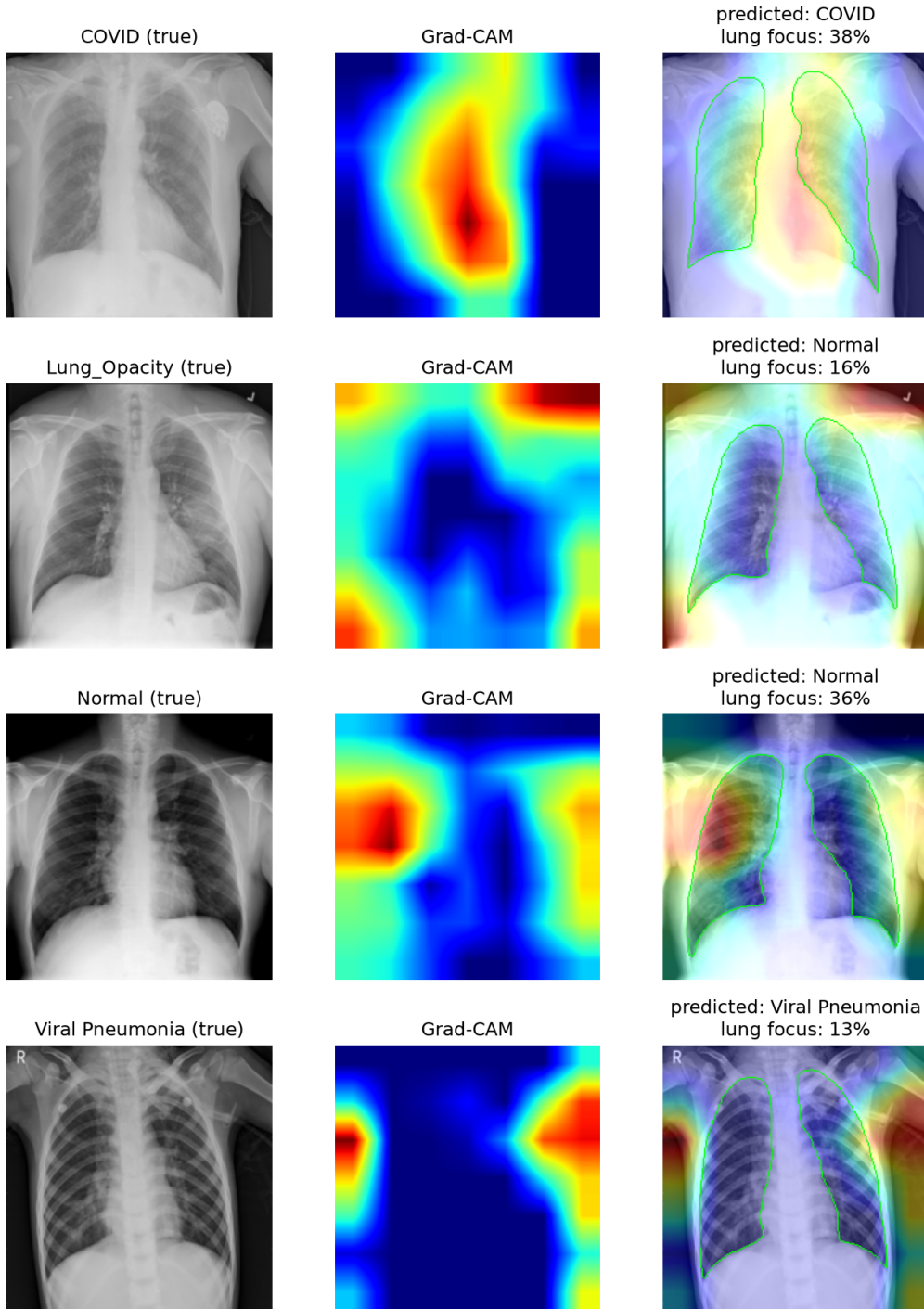


Figure 5.3: Representative Grad-CAM examples for the frozen transfer model, with the lung mask boundary overlaid in green.

Table 5.5 quantifies this over 300 sampled test images.

Table 5.5: Grad-CAM lung focus versus chance for the frozen transfer model.

Class	Grad-CAM in Lungs	Chance Level	Difference
COVID	30.8%	24.7%	+6.1 pts
Normal	28.5%	25.0%	+3.5 pts
Viral Pneumonia	29.8%	25.7%	+4.2 pts
Lung Opacity	17.7%	20.6%	−2.9 pts

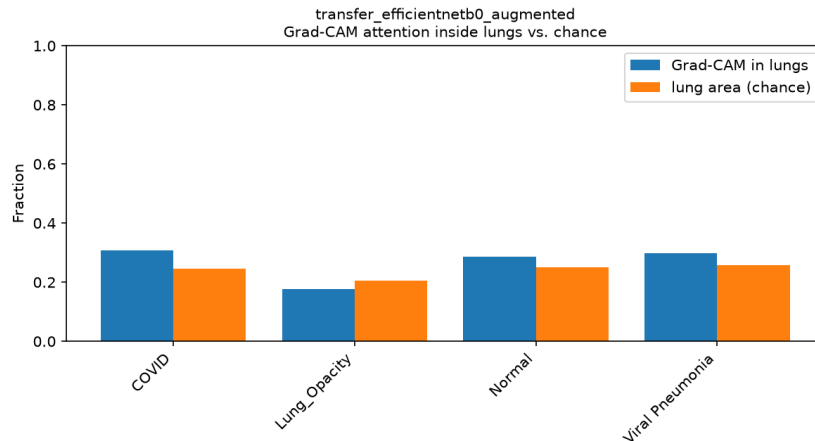


Figure 5.4: Grad-CAM attention inside the lungs versus the chance level, by class, for the frozen transfer model.

Attention is only mildly above chance for three of the four classes and below chance for Lung Opacity, meaning the model attends to non-lung regions more than pure geometry would predict for that class. Splitting the same sample by prediction outcome showed virtually identical lung focus for correct predictions (25.4% versus a chance level of 23.7%) and misclassified predictions (25.2% versus 22.5%), indicating that weak lung-focused attention is a systemic property of this model rather than an error-specific artefact.

5.5.2 Causal Test: Masking the Input

Correlational Grad-CAM evidence cannot by itself establish that the model’s accuracy depends on non-pulmonary cues. A direct causal test was therefore performed by training an otherwise identical model on images with every pixel outside the lung mask forcibly set to zero before reaching the network, so that no background information was available at all.

Table 5.6 compares this masked model against the frozen baseline.

Table 5.6: Effect of masking the background at the input (causal test), test split.

Metric	Full Image	Lungs Only (Masked)	Change
Test accuracy	89.70%	83.07%	−6.6 pts
Test macro F1	0.907	0.823	−8.4 pts
COVID recall	90.3%	57.0%	−33.3 pts
COVID F1	0.923	0.678	−24.5 pts
Lung Opacity F1	0.854	0.809	−4.5 pts
Normal F1	0.906	0.872	−3.4 pts
Viral Pneumonia F1	0.945	0.932	−1.3 pts

If the model’s accuracy depended evenly on pulmonary pathology, removing the background should have degraded every class by a comparable amount. Instead, the loss is heavily concentrated in COVID: recall collapses from 90.3% to 57.0%, a 33.3-point drop, meaning the masked model misses more than 40% of COVID cases that the full-image model correctly identified. The other three classes lose only a few points each. This pattern is direct causal evidence of COVID-specific background leakage in this dataset, consistent with the widely reported concern that COVID images in this Kaggle collection were aggregated from different sources than the other classes, introducing incidental correlations unrelated to disease status.

As expected, Grad-CAM attention moves substantially into the lungs once the background is physically removed, since no other information remains available to the network (Table 5.7, Figure 5.5).

Table 5.7: Grad-CAM lung focus versus chance for the masked transfer model.

Class	Grad-CAM in Lungs	Chance Level	Difference
COVID	40.8%	24.7%	+16.1 pts
Lung Opacity	30.2%	20.6%	+9.6 pts
Normal	49.4%	25.0%	+24.4 pts
Viral Pneumonia	44.4%	25.7%	+18.7 pts

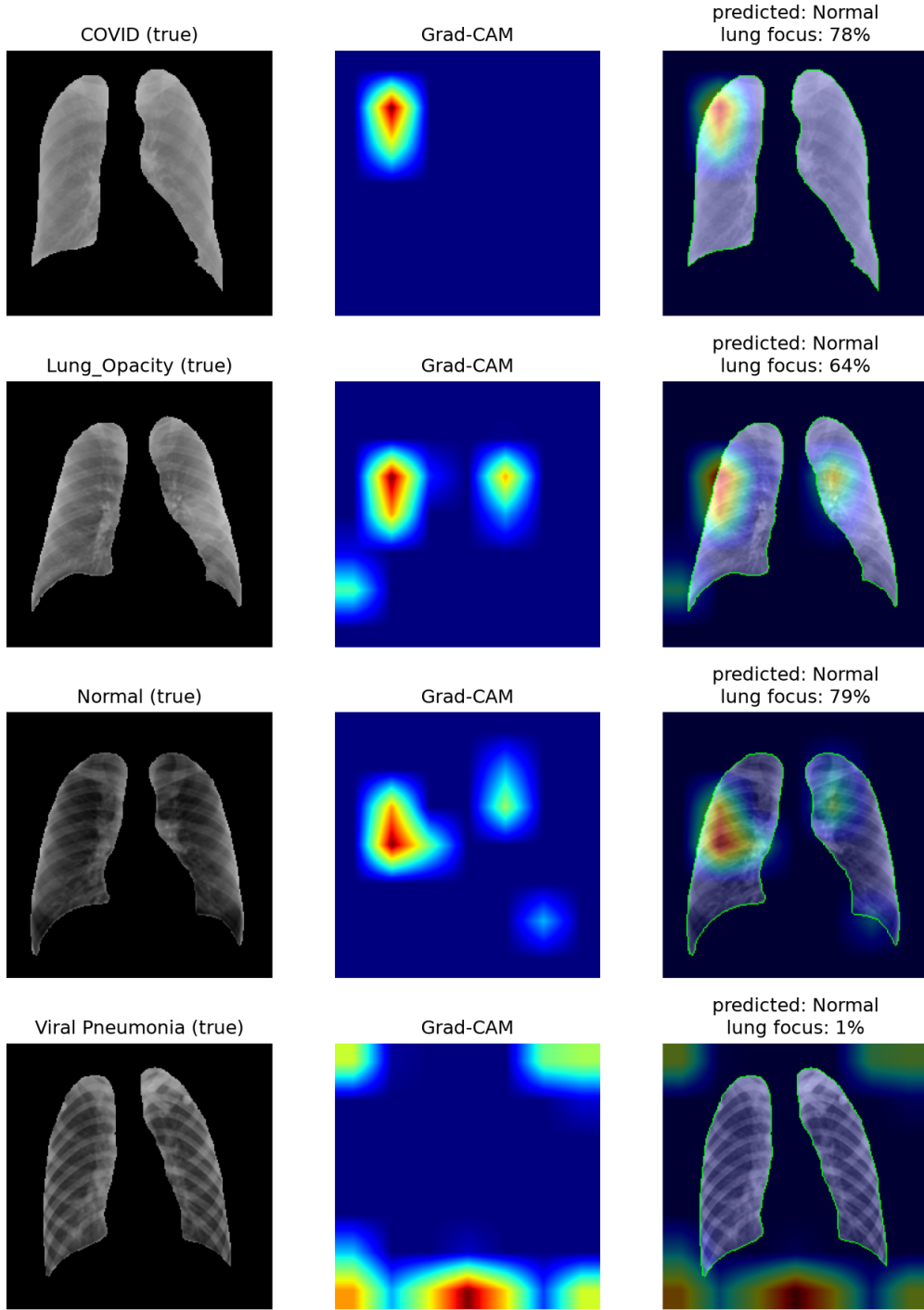


Figure 5.5: Representative Grad-CAM examples for the masked transfer model, with the lung mask boundary overlaid in green. All model input outside the boundary was zeroed before training and inference.

A small number of heatmap regions still extend outside the lung boundary even though the corresponding input pixels were exactly zero. This is not evidence that the masking failed; it is a known limitation of Grad-CAM on deep convolutional networks, caused by non-zero learned biases and batch normalisation statistics propagating through many stacked layers, together with the coarse spatial resolution of the final convolutional feature map and its associated large receptive field. For this reason, the aggregated lung-

focus statistics, and above all the masked-training accuracy experiment itself, are treated as the primary evidence, rather than any single Grad-CAM pixel.

5.5.3 Isolating Shape: A Mask-Only Model

The masked model in Section 5.5.2 still receives the lung’s pixel intensities, that is, its texture and density patterns, and only removes the background. To isolate lung *shape* from lung *texture*, a further model was trained on the segmentation masks themselves, that is, a binary lung silhouette with no pixel-intensity information from the original radiograph at all.

Table 5.8 extends the comparison to this shape-only model.

Table 5.8: Decomposing accuracy into background, texture, and shape contributions (test split).

Metric	Full Image	Lungs, Texture Kept	Shape Only
Test accuracy	89.70%	83.07%	74.07%
Test macro F1	0.907	0.823	0.703
COVID recall	90.3%	57.0%	29.3%
COVID F1	0.923	0.678	0.413
Normal F1	0.906	0.872	0.818
Lung Opacity F1	0.854	0.809	0.709
Viral Pneumonia F1	0.945	0.932	0.873

Lung shape alone is a weak classifier for COVID, with recall of 29.3%, close to what a random four-class guess would achieve, while Normal cases remain comparatively easier to recognise from shape alone (F1 0.818). The test confusion matrix (Figure 5.6) shows that, of the true COVID cases, the shape-only model most often mistakes them for Normal or Lung Opacity, consistent with COVID’s radiographic signature being primarily about tissue density and opacity patterns inside the lung field rather than its outline.

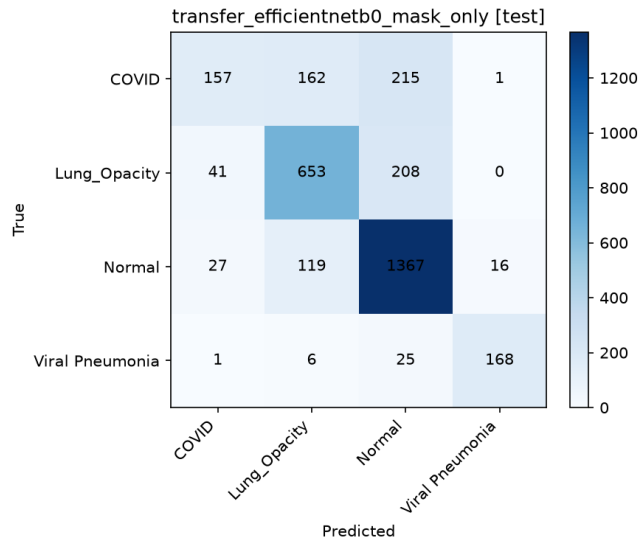


Figure 5.6: Test-set confusion matrix for the shape-only (mask-only) transfer model.

Taken together, Sections 5.5.2 and 5.5.3 give a three-way decomposition of the frozen model’s 89.70% test accuracy: approximately 6.6 points depend on non-lung background and context, a further approximately 9.0 points depend on lung texture and density patterns rather than shape, and the remainder is what lung geometry alone can achieve, well above the 25% random baseline but a weak classifier by itself, and specifically unreliable for COVID.

5.5.4 An Architectural Alternative: Masked Pooling

Hard input masking (Section 5.5.2) is an effective causal probe, but it discards all information outside the lungs even where that information might legitimately help, for example by providing context for image quality or overall framing, and it costs several points of raw performance. As a softer alternative, a masked global average pooling layer was used instead: rather than zeroing the input image, the model’s final convolutional feature map is masked so that only lung-covering spatial locations contribute to the pooled features that feed the classification head, while the backbone still observes the full image.

This configuration, combined with class balancing and rotation-only augmentation, reached **88.79%** test accuracy and **0.897** macro F1, recovering to within roughly one accuracy point of the frozen full-image baseline while still architecturally constraining the features used for classification to the lung region. This suggests that how a model is restricted to the lungs, that is, at the input versus within the pooling operation, materially affects the performance cost of enforcing that constraint, and is a useful direction for future work aiming to reduce background reliance without the full accuracy cost of hard input masking.

5.5.5 Grad-CAM on the Fine-Tuned Model

The same lung-focus analysis was repeated for the fine-tuned model of Section 5.4, the best-performing model in this project. Table 5.9 reports the per-class results.

Table 5.9: Grad-CAM lung focus versus chance for the fine-tuned transfer model.

Class	Grad-CAM in Lungs	Chance Level	Difference
COVID	24.3%	24.4%	−0.1 pts
Lung Opacity	31.2%	20.9%	+10.3 pts
Normal	38.5%	25.2%	+13.3 pts
Viral Pneumonia	23.8%	24.0%	−0.2 pts

Fine-tuning changes which classes show above-chance lung attention: Lung Opacity and Normal now show clearly above-chance attention, the opposite of the frozen baseline’s pattern in Table 5.5, while COVID and Viral Pneumonia sit almost exactly at chance. This differs from the CNN-from-scratch full-image baseline’s attention pattern (Section 4.2.1), where COVID and Normal were close to proportional and Lung Opacity was the weakest. The two architectures therefore appear to rely on different, model-specific combinations of pulmonary and non-pulmonary cues, even though both ultimately fall short of attention that is consistently and exclusively pulmonary. Given the masking experiments in Section 5.5.2, the near-chance COVID attention of the fine-tuned model is a reminder that its very high COVID recall (97.8%, Table 5.4) should not be assumed

to reflect improved pulmonary diagnosis without further, dedicated causal testing on this specific model.

5.6 Comparison with the Simple CNN Baselines

Table 5.10 brings together the Simple CNN baselines of Chapter 4 and the transfer-learning models of this chapter on the common test split.

Table 5.10: Comparison of all principal models evaluated in this project (test split).

Model	Test Accuracy	Macro F1
Simple CNN, full image	81.96%	0.827
Simple CNN, lung-ROI	78.71%	0.774
Transfer, frozen EfficientNetB0	89.70%	0.907
Transfer, augmented (no flip)	88.38%	0.896
Transfer, fine-tuned	92.36%	0.936

Figure 5.7 shows the same comparison extended to every transfer-learning run trained in this chapter.

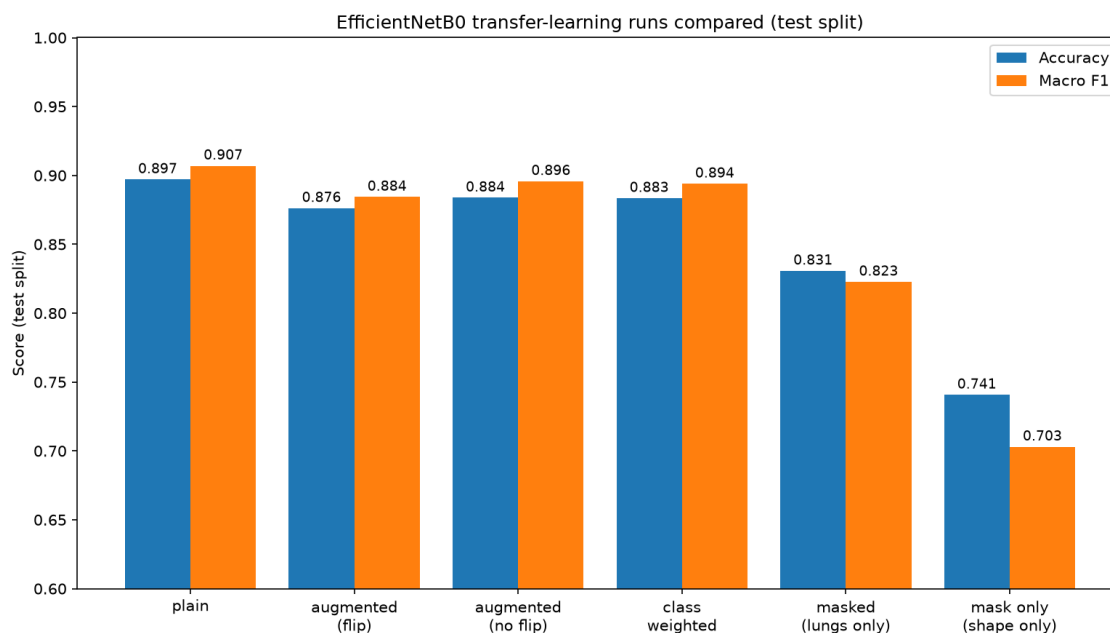


Figure 5.7: Test accuracy and macro F1 across all EfficientNetB0 transfer-learning runs compared in this chapter.

Three conclusions follow from this chapter, in line with the project’s Step 3 objective of drawing scientific and business conclusions from the success or failure of the modelling phase.

First, transfer learning substantially outperforms training a comparable CNN from scratch on this dataset, and fine-tuning the backbone end-to-end gives the strongest result obtained anywhere in this project (92.36% accuracy, 0.936 macro F1). This is consistent

with the dataset being too small, relative to the visual complexity of chest radiographs, for a from-scratch model to learn as effective a feature representation as one pretrained on a much larger, general-purpose image corpus.

Second, this headline performance is not fully attributable to pulmonary pathology. The interpretability analysis in Section 5.5 shows that a meaningful share of the frozen model’s accuracy, and specifically its COVID recall, depends on non-anatomical cues in this Kaggle dataset rather than lung tissue itself. The fine-tuned model’s near-chance COVID attention (Section 5.5.5) suggests this concern is not resolved simply by improving raw accuracy through fine-tuning, and may require dedicated causal testing analogous to Section 5.5.2 before it can be ruled out.

Third, and consequently, the headline accuracy of the best model in this project should not be read as evidence of clinical readiness. As with the CNN-from-scratch baselines, this remains an educational decision-support study rather than a validated diagnostic system, and any future external validation of this model line should specifically test for COVID-recall collapse on radiographs from sources not represented in this training dataset, since that is precisely the failure mode predicted by the background-leakage evidence in this chapter.

Chapter 6

Analysis: What are the models actually learning?

Chapters 4 and 5 reported classification performance and, in each case, interpreted a single model. Section 5.5.2 in particular established that removing the background from the input causes the frozen EfficientNetB0 model to lose 33.3 points of COVID recall, which is direct causal evidence that its accuracy depends on non-pulmonary information.

That result concerns one architecture. This chapter asks whether it is a property of that model or a property of the dataset. Four independent lines of evidence are presented: the source composition of the dataset itself, a region ablation replicated across every architecture trained in this project, a resolution ablation, and a combined condition in which both anatomical detail and the lung field are removed simultaneously. Each addresses an objection that the others leave open.

6.1 Source Composition of the Dataset

Table 2.1 noted that the four diagnostic classes were drawn from different repositories. The metadata distributed with the dataset allows this to be quantified exactly, since every image records the URL of its originating collection.

Table 6.1 cross-tabulates source repository against diagnostic label.

Table 6.1: Number of images contributed by each source repository to each diagnostic class.

Source repository	COVID-19	Lung Opacity	Normal	Viral Pneumonia
BIMCV	2,473	0	0	0
ml-workgroup	183	0	0	0
Eurorad	559	0	0	0
SIRM	94	0	0	0
ieee8023	183	0	0	0
COVID-CXNet	124	0	0	0
RSNA Pneumonia Challenge	0	6,012	8,851	0
Kaggle Chest X-ray	0	0	1,341	1,345

The structure of this table is the central observation of this chapter. Six repositories contribute COVID-19 images and nothing else; no COVID-19 image originates from either

of the two repositories that supply the remaining three classes. Source membership therefore identifies COVID-19 with certainty. The two remaining sources are also partially informative: the RSNA collection contributes only Lung Opacity and Normal cases, and the Kaggle collection only Viral Pneumonia and Normal cases.

To express this as a performance figure, a trivial classifier was constructed that ignores the image entirely and predicts, for each source, that source’s majority class. Its results are given in Table 6.2.

Table 6.2: Performance of a classifier that uses only the source repository and no image data whatsoever.

Metric	Value
Accuracy	65.26%
Macro F1	0.594
COVID-19 F1	1.000
Normal F1	0.707
Viral Pneumonia F1	0.667
Lung Opacity F1	0.000

Knowing only which repository an image came from, and nothing about its content, yields perfect COVID-19 classification and 65.26% overall accuracy. Measured as mutual information, source membership removes 52.6% of the total label uncertainty ($I = 0.625$ nats against a label entropy of $H = 1.186$ nats).

This does not by itself demonstrate that any trained model exploits the correspondence. It establishes the weaker but necessary precondition: the information is present, it is strong, and it is concentrated precisely in the class whose detection is the stated objective. Whether a model can recover source membership from the pixels is the question addressed by the remaining sections.

A second confound is worth recording here. All Viral Pneumonia images derive from a single paediatric collection, whereas the other three classes are predominantly adult. Differences in body proportion between children and adults are visible throughout the radiograph and are not confined to the lung field.

6.2 Region Ablation Across Architectures

Section 5.5.2 restricted the EfficientNetB0 input to the segmented lung field. The complementary experiment restricts the input to everything *except* the lung field, by inverting the same mask. Both conditions zero the excluded pixels rather than removing them, so the two inputs have identical dimensions and a given pixel position carries the same meaning in each.

This pair of conditions was run for every architecture trained in this project. Results are reported in Table 6.3.

The ordering is consistent without exception. In all four conditions, a model trained exclusively on the pixels outside the lung field outperforms the same model trained exclusively on the lung field. The effect spans architectures differing by more than two orders of magnitude in capacity, from a 23,556-parameter CNN trained from scratch to

Table 6.3: Test macro F1 by input region, across architectures. The final column is the difference between the background-only and lungs-only conditions; a positive value indicates that the non-lung pixels are more informative than the lung field.

Model	Input	Full	Lungs	Background	Bg – Lungs
EfficientNetB0 (frozen)	224 ²	0.907	0.823	0.876	+0.053
Simple CNN	128 ²	0.827	0.690	0.741	+0.051
Simple CNN	16 ²	0.835	0.749	0.787	+0.038
LeNet-5	32 ²	0.815	0.739	0.747	+0.008

a pretrained ImageNet backbone of approximately four million parameters, and it does not depend on input resolution.

For EfficientNetB0 the background-only model reaches 0.876 macro F1 against 0.907 for the unrestricted model. Deleting the entire lung field costs 0.031 macro F1 points.

That the lung field carries *less* class information than its complement is difficult to reconcile with any account in which these models detect pulmonary pathology.

The region ablation in Table 6.3 uses frozen-backbone models. Full-backbone fine-tuning at 224×224 exhausted available memory on the hardware used for this project, so the background-only condition was not repeated under fine-tuning. Section 6.5 provides partial evidence that the effect persists after fine-tuning, since the lung-ROI restriction costs a comparable amount in both settings.

6.3 Resolution Ablation

A remaining objection is that the background regions might carry a coarse correlate of pathology, for example through overall image intensity, rather than a source signature. If so, the models would still be using disease-related information, merely in an indirect form.

The simple CNN was therefore trained at three input resolutions, with all other settings held constant. Table 6.4 reports the results.

Table 6.4: Test performance of the simple CNN as a function of input resolution, full-image input.

Input resolution	Pixels	Macro F1
224×224	50,176	0.791
128×128	16,384	0.827
16×16	256	0.835

Performance does not degrade as resolution falls. The 16×16 model, receiving 196 times fewer pixels than the 224×224 model, scores slightly higher than both larger inputs.

At 16×16 the thorax occupies approximately 10×10 pixels and the lung fields roughly 8×8 . Radiographic findings such as ground-glass opacity, consolidation, or interstitial infiltrate are not resolvable at that scale. Whatever the network uses to separate the four classes survives the destruction of essentially all anatomical detail, which points to coarse global image properties rather than pathology.

6.4 Combined Condition: Low Resolution Without the Lung Field

Sections 6.2 and 6.3 each leave one route open. The region ablation permits the objection that the background model exploits coarse anatomical spill-over; the resolution ablation permits the objection that coarse whole-lung opacity survives downsampling. Applying both restrictions simultaneously closes both.

Table 6.5 reports the three region conditions at 16×16 , with identical architecture, resolution, random seed, and training configuration. The only difference between rows is which pixels the model is permitted to see.

Table 6.5: Simple CNN at 16×16 input under three region conditions (test split). All other settings are identical across rows.

Region	Accuracy	Macro F1	COVID F1
Full image	83.70%	0.835	0.778
Background only	78.74%	0.787	0.669
Lungs only	75.93%	0.749	0.532

A model with 23,556 parameters, seeing a 16×16 thumbnail with the lung field deleted, achieves 0.787 macro F1 against a majority-class baseline of 0.163. Removing the lungs costs 0.048 macro F1; removing the background costs 0.086. The lung field is the least informative of the three inputs.

The class-wise figures are more pointed than the aggregate. COVID-19 reaches F1 0.669 from the background alone but only 0.532 from the lung field. The class whose detection motivates the entire task is the one the lung field handles worst and the surrounding pixels handle best. This is precisely the pattern predicted by Section 6.1, where COVID-19 was shown to be perfectly separable by source membership.

6.5 Region Restriction Under a Controlled Comparison

The lung-ROI baseline of Section 4.3 and the transfer-learning ROI experiments of Chapter 5 used different crop definitions, which prevents a direct comparison between the two model families. To remove this confound, the transfer-learning pipeline was extended to use the same square lung-ROI extraction as the simple CNN baselines, producing byte-identical inputs for both architectures.

Under matched fine-tuning settings, EfficientNetB0 reaches 0.903¹ macro F1 on full images and 0.889 on the lung ROI: a cost of 0.014 for discarding roughly two thirds of the image. The corresponding simple CNN figures are 0.827 and 0.774.

The architecture gap of approximately 0.08 macro F1 is therefore attributable to pretraining rather than to differences in input preparation. The region effect, however, is of the same sign and comparable magnitude in both.

¹These runs use a reduced fine-tuning configuration (i.e., 20 unfrozen backbone layers and a batch size of 16) rather than the full-unfreeze configuration of Section 5.4, owing to memory constraints. Both conditions use identical settings, so the difference between them is attributable to the input region alone, although neither reaches the peak performance reported in Chapter 5.

6.6 Synthesis

The four experiments answer different objections and converge on one conclusion.

1. **The information exists.** Source membership determines the COVID-19 label exactly and removes 52.6% of total label uncertainty, before any model is trained.
2. **Models recover it.** Every architecture tested performs better on the non-lung pixels than on the lung field.
3. **It requires no anatomical detail.** Performance at 16×16 matches performance at 224×224 .
4. **Neither route is necessary.** At 16×16 with the lung field removed, macro F1 remains at 0.787.

Two qualifications are warranted. First, these results do not show that the lung field carries no diagnostic signal: restricting models to the lungs leaves performance well above chance, and Section 5.5.3 showed that lung texture contributes approximately nine accuracy points beyond lung shape alone. The claim is that the extra-pulmonary signal is sufficient on its own, not that the pulmonary signal is absent.

Second, the effect is not an artefact of weak models. Section 5.5.2 demonstrated it in the frozen EfficientNetB0 model at 89.70% accuracy, which is the strongest frozen-backbone configuration evaluated in this project, and Section 6.2 reproduces it in every architecture tested. Capacity does not protect against the shortcut: the pretrained backbone shows the largest background-minus-lungs margin of any model in Table 6.3.

The headline figures reported in Chapters 4 and 5, including the 92.36% accuracy of the fine-tuned model, are consistent with published results on the same dataset and task: El Houby (2024) report 0.935 accuracy for four-class classification. The experiments above indicate that such figures cannot be read as measurements of diagnostic ability. A substantial share of the achievable performance is available from source-identifying characteristics of the images, and for COVID-19 the correspondence between source and label within this dataset is exact. That correspondence carries no information outside it.

This corroborates the findings of DeGrave, Janizek, and Lee 2021 on different COVID-19 radiograph collections, and extends them in two respects: the effect is shown here to hold across three architectures and two input resolutions, from a 23,556-parameter network trained from scratch to a pretrained ImageNet backbone, and it is traced to a documented source-label correspondence in the dataset metadata rather than inferred from model behaviour alone.

Chapter 7

Conclusion

7.1 Summary of Findings

This project evaluated convolutional neural networks for four-class classification of chest radiographs into COVID-19, Lung Opacity, Normal, and Viral Pneumonia, and examined what those networks use to make their predictions.

We address two questions, posed in Section 1.3:

1. **Performance.** What classification accuracy can convolutional neural networks achieve on this dataset, and how much does transfer learning from a pretrained backbone improve on a network trained from scratch?
2. **Attribution.** Which regions and properties of the input do these models rely on, and to what extent is their accuracy attributable to pulmonary pathology rather than to characteristics of the source repositories?

On the first question, the results follow a familiar pattern. A small convolutional network trained from scratch reached 81.96% test accuracy and 0.827 macro F1. Transfer learning from an ImageNet-pretrained EfficientNetB0 backbone improved this substantially without any fine-tuning, to 89.70% accuracy and 0.907 macro F1, and fine-tuning the backbone end-to-end produced the strongest result obtained anywhere in this project at 92.36% accuracy and 0.936 macro F1. These figures are consistent with published work on the same dataset and task.

On the second question, the results are less reassuring. Four independent lines of evidence, presented in Chapter 6, indicate that a substantial share of this performance does not derive from pulmonary pathology:

- The dataset metadata records the originating repository of every image. Six repositories contribute COVID-19 cases and nothing else, so source membership identifies COVID-19 exactly. A classifier using only the source label and no image data reaches 65.26% accuracy and perfect COVID-19 F1.
- Every architecture evaluated performs better when trained on the pixels outside the segmented lung field than on the lung field itself. For the frozen EfficientNetB0 model, training on the background alone gives 0.876 macro F1, compared with 0.823 when trained on the lungs alone. Removing the lungs entirely costs only 0.031 relative to the full image.

- Reducing the input resolution from 224×224 to 16×16 does not degrade performance, although radiographic findings are not resolvable at that scale.
- Combining both restrictions, a 23,556-parameter network seeing a 16×16 thumbnail with the lung field removed still reaches 0.787 macro F1.

Grad-CAM analysis is consistent with this. Attention fell close to or below the chance level defined by lung area for several classes, in both the from-scratch and transfer models, and the causal masking experiment in Section 5.5.2 showed COVID recall collapsing by 33.3 points when background pixels were removed.

The central conclusion is therefore not about which architecture performs best. It is that accuracy on this dataset is a poor proxy for diagnostic ability, because the dataset’s construction makes source identification an effective substitute for the intended task. These findings agree with those of DeGrave, Janizek, and Lee (2021), who identified the same form of shortcut learning in other COVID-19 radiograph datasets. The present work adds two things: it traces the shortcut to a source-label correspondence documented in the dataset metadata, and it shows that the effect persists when input resolution is reduced far below the scale at which radiographic findings are visible.

7.2 Limitations

Several limitations constrain the scope of these conclusions.

No external validation. All results are reported on a held-out split of a single dataset. The strongest test of the shortcut hypothesis would be evaluation on radiographs from institutions not represented here, which would be expected to show a substantial performance drop, particularly for COVID-19. No such external dataset was used in this project, so the predicted failure is inferred rather than observed.

Hardware constraints on fine-tuning. Full-backbone fine-tuning at 224×224 exhausted available memory on the hardware used. The region ablation in Table 6.3 therefore uses frozen-backbone models, and the matched region comparison in Section 6.5 uses a reduced fine-tuning configuration. Whether the background-only condition would behave differently under full fine-tuning was not tested.

Single training run per condition. Each configuration was trained once with a fixed random seed. Run-to-run variance was not quantified, so small differences between conditions should be treated as indicative rather than established. The principal findings rest on differences large enough that this is unlikely to affect them, but the point applies to the closer comparisons, such as the 0.014 gap between the full-image and lung-ROI fine-tuned models.

Segmentation masks as ground truth. The lung masks distributed with the dataset were themselves produced by a segmentation model. Errors in those masks propagate into every region-restricted experiment. A systematically over-inclusive mask would move genuine lung tissue into the “background” condition, and a systematically under-inclusive one would do the reverse. Mask quality was not independently assessed.

Near-duplicate images. Exact duplicates were removed before splitting, but 196 image pairs sharing a perceptual hash while differing at the pixel level were retained, as automatic exclusion risked discarding legitimately distinct acquisitions. Any residual leakage from near-duplicates would inflate all reported figures equally and does not affect the comparisons between conditions.

Class scope. The four-class formulation used here is harder than the three-class variant common in the literature, which omits Lung Opacity. Comparisons with published accuracies should account for this.

7.3 Future Work

External validation. The most informative next step would be to evaluate the trained models on a radiograph collection from institutions absent from this dataset. The shortcut hypothesis makes a specific, falsifiable prediction: COVID-19 recall should degrade far more than the other classes.

Predicting source directly. A useful diagnostic would be to train a classifier whose target is the source repository rather than the diagnosis. Its accuracy would quantify directly how legible source identity is from the pixels, and comparing its Grad-CAM attention with that of the diagnostic models would show whether both rely on the same image regions.

Testing architectural restrictions. The masked-pooling model in Section 5.5.4 constrains the final pooled features to lung-positioned locations and loses only about one accuracy point relative to the full-image baseline, compared with 6.6 points for input-level masking. This should not be read as evidence that the model uses only pulmonary information. In a deep network the features at a given location summarise a receptive field covering much of the image, so lung-positioned features may still encode background content. Testing this directly—for example, by evaluating the masked-pooling model on images whose non-lung regions have been replaced or perturbed—would establish whether architectural restriction removes the shortcut or merely conceals it.

Dataset construction. The most durable fix is not a modelling one. A dataset in which each diagnostic class draws from multiple institutions, and each institution contributes multiple classes, would break the source-label correspondence at its origin. Reporting per-source performance would allow readers to detect the problem where it remains.

7.4 Closing Remarks

The best model in this project reached 0.936 macro F1 and 92.36% test accuracy, in line with published results on this dataset. The evidence presented here indicates that these figures cannot be interpreted as measures of diagnostic ability. Grad-CAM analysis in Chapters 4 and 5 measured how much of each model’s attention fell inside the lungs, compared with how much would fall there by chance given the lungs’ share of the image.

For most classes the two were close, and for some attention inside the lungs was lower than chance. For the fine-tuned model classifying COVID-19, attention inside the lungs was 24.3%, against a chance level of 24.4%: the model attended to the lungs no more than it would have by chance.

The ablations in Chapter 6 then showed that a substantial share of the achievable performance is available from characteristics of the source repositories rather than from the lungs. Establishing where a model’s performance comes from therefore needs to be treated as part of its evaluation, not as an optional addition to it.

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